

The tetrahedral Horn problem and asymptotics of $U(n)$ $6j$ symbols

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Abstract

For a compact reductive group G , Wigner $6j$ -symbols are maps between multiplicity spaces which encode the associativity constraint $(V_\alpha \otimes V_\beta) \otimes V_\delta \cong V_\alpha \otimes (V_\beta \otimes V_\delta)$ for unitary irreducible representations of G . For $G = SU(2)$, their asymptotic behavior for large weights was conjectured by Ponzano-Regge and established by Roberts using geometric quantization techniques.

In this paper, we address the problem of asymptotic behavior of $6j$ -symbols for $G = U(n)$. We show that their decay rate is polynomial or exponential depending on the existence of a tetrahedron in the space of Hermitian n by n matrices with edges determined by the 6 arguments of the $6j$ -symbol. Our proof is inspired by probabilistic techniques from quantum information theory.

As an application of our results, we obtain a new proof of the strongly subadditive entropy inequality for tetrahedra in the space of Hermitian matrices (the tetrahedral Horn problem).

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1 Introduction

Horn’s problem: The problem of determining the set of eigenvalues (a, b, c) of Hermitian matrices (A, B, C) satisfying $A + B = C$ has been investigated for over a century [Wey12; Fan49; Lid53; Wie55; TF71; TF70] and has come to be known as *Horn’s problem* after Horn who, in 1962, conjectured a solution to the problem in the form of an inductively defined, finite set of homogeneous linear inequalities. In 1998, Klyachko provided a solution to Horn’s problem [Kly98] which, when combined with a proof of the so-called saturation conjecture, due to Knutson and Tao [KT99], constitutes a proof of Horn’s original conjecture [Ful98]. These linear inequalities, now known as the *Horn inequalities* describe the convex polytope of admissible values for (a, b, c) . Recent developments on Horn’s problem include the development of the geometric method by Sherman [She15] and Berline, Vergne, and Walter [BVW18] originally pioneered by the geometric proofs due to Belkale [Bel02; Bel05], as well the characterization and computation of the induced distribution over the eigenvalues γ of the sum $C = A + B$ if A and B are sampled uniformly and independently from the respective sets of Hermitian matrices with eigenvalues a and b by Coquereaux, McSwiggen, and Zuber in series of papers [CZ18; CZ18; CMZ19; Zub18]. For a pedagogical and historical introduction to Horn’s problem, we recommend the review articles of Bhatia [Bha01], Knutson [Knu00], and Fulton [Ful00] or alternatively [NP06, Appendix D]. See also the article by Daftuar and Hayden for an account of applications to topics in quantum information theory [DH05].

In parallel to Horn’s problem, numerous variations and extensions of Horn’s problem have been explored by various authors. For example, numerous authors have considered the admissible eigenvalues of unitary matrices and their product [AW97; Bel08; BK16], or the admissible singular values of matrices and their product [Spe05; TT71]. In 2000, it was shown by [Kly00] that *logarithms* of the admissible singular values of non-singular matrices and their product are identical to the admissible eigenvalues of Hermitian matrices and their sum, thus proving a conjecture due to Thompson [Kly00; AMW01].

The tetrahedral Horn problem: In this paper, we consider a generalization of Horn’s problem investigated previously only briefly in [CSW18]. While Horn’s problem aims to characterize the eigenvalues (a, b, c) of a triple of $n \times n$ Hermitian matrices (A, B, C) which are subjected to the constraint $A + B = C$, the generalization we consider here aims to characterize the possible eigenvalues (a, b, c, d, e, f) of a *sextuple* of $n \times n$ matrices (A, B, C, D, E, F) subject to four constraints:

$$A + B = C, \quad B + D = F, \quad D + C = E, \quad \text{and} \quad A + F = E. \quad \text{\texttt{\{eq:tet_cond\}}} \quad (1)$$

These constraints arise naturally from building a tetrahedral arrangement of $n \times n$ Hermitian matrices inside the vector space of such Hermitian matrices with edges labelled according to Fig. 1. For this reason, we refer to this eigenvalue problem as the *tetrahedral* Horn problem and we say a sextuple of eigenvalues, (a, b, c, d, e, f) , is *tetrahedral* if such matrices exist, and not tetrahedral otherwise.

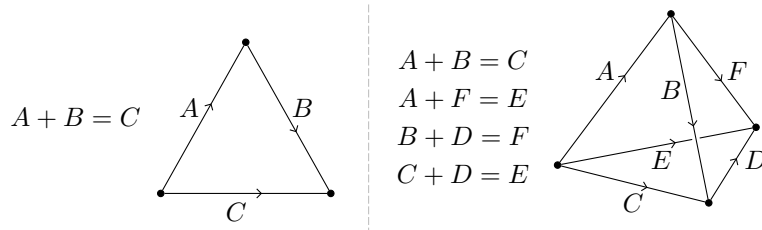


Figure 1: (Left) Horn’s problem subjects three Hermitian matrices, (A, B, C) , to the constraint $A + B = C$ forming a triangle in the vector space of Hermitian matrices. (Right) The *tetrahedral* Horn problem subjects six Hermitian matrices, (A, B, C, D, E, F) , to a quadruple of constraints forming a tetrahedron in the vector space of Hermitian matrices. fig:tetrahedron_labelling_scheme

Beyond the superficial connection between the imposed constraints and the resulting shape formed in the vector space of Hermitian matrices (as in Fig. 1), in the special case of 2×2 Hermitian matrices, this connection is much deeper. It is well-known that the space of admissible eigenvalues for a triangular arrangement

of 2×2 traceless Hermitian matrices is equivalent to the moduli space of Euclidean *triangles* in $\mathbb{R}^3$ [HK96; FKL10; Knu00]. This correspondence generalizes to the tetrahedral case in the sense that the space of admissible eigenvalues for a tetrahedral arrangements of 2×2 traceless Hermitian matrices is equivalent to the moduli space of Euclidean *tetrahedra* in $\mathbb{R}^3$. While these correspondences are straightforward to prove and also well-known, this geometric knowledge about the solution space leads to a complete solution to the tetrahedral Horn's problem for 2×2 matrices; specifically, the moduli space of Euclidean tetrahedra in $\mathbb{R}^3$ is described by the conjunction of the triangle inequalities for each of the four faces (these inequalities coincide with the solution to Horn's original problem) together with the positivity of so-called Cayley-Menger determinants which are polynomial inequalities in the side-lengths (a, b, c, d, e, f) [SJS18]. Even for $n = 2$, we remark that there is a strong qualitative difference between the tetrahedral Horn problem and the original Horn problem. In particular, while the Horn's problem is completely characterized by a *finite* number of *linear* inequalities in the eigenvalues (indeed the solution space is an instance of a moment polytope known as *Horn's polytope*), the solution space for the tetrahedral Horn's problem is not convex nor a polytope.

Previous work: TC: Add a summary of all the strong subadditivity stuff. More recently, Alekseev et al. have investigated a few variants of the tetrahedral Horn problem concerning either the singular values of products of invertible matrices or the tropical singular values of concatenations of weighted planar networks [Ale+25]. These variants are termed *multiple* Horn problems in [Ale+25]; the tetrahedral Horn problem we consider here is termed the *additive* multiple Horn problem in the language of [Ale+25].

Our contributions: Regarding the tetrahedral Horn problem, we derive a new family of inequalities constraining the set of tetrahedral eigenvalues. These inequalities are expressed in terms of Schur polynomials $s_\lambda(x)$ in the eigenvalues $x = (x_1, \dots, x_n)$ which are homogeneous symmetric polynomials indexed by so-called Young diagrams λ . More accurately, we find our inequalities are best articulated in terms of a simple rescaling of the Schur polynomials $\varphi_\lambda(x) = s_\lambda(x)/H_\lambda$ where H_λ is a positive integer equal to the product of the hook-lengths of all boxes in the Young diagram λ . These inequalities are presented in the following theorem. thm:nec_tet

Theorem 1. *If (a, b, c, d, e, f) are the eigenvalues of positive semidefinite $n \times n$ matrices (A, B, C, D, E, F) satisfying $A + B = C$, $B + D = F$, $C + D = E$ and $A + F = E$, then for all triples of Young diagrams (α, β, δ) ,*

$$\varphi_\alpha(a)^{\frac{2}{3}} \varphi_\beta(b) \varphi_\delta(d)^{\frac{2}{3}} \leq \sum_{\gamma, \epsilon, \phi} \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_\infty \varphi_\phi(f)^{\frac{1}{3}} \varphi_\gamma(c)^{\frac{1}{3}} \varphi_\epsilon(e)^{\frac{1}{3}}. \quad (2) \quad \text{[eq:nec_tet]}$$

The expression $\begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix}$ appearing here is known as the $6j$ -symbol for the group of unitary $n \times n$ matrices $U(n)$ and its significance will be explained shortly. Moreover, we quantify the utility of these inequalities by proving the following distance bound. thm:suff_tet

Theorem 2. *If (a, b, c, d, e, f) are non-negative eigenvalues satisfying the trace conditions and also Eq. (2) for all triples (α, β, δ) such that $|\alpha| + |\beta| + |\delta| = k$ for some positive integer k , then*

$$\frac{1}{\text{Tr}[e]} D \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \leq O \left(n \sqrt{\frac{\log k}{k}} \right), \quad (3)$$

where $D \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \geq 0$ with equality holding if and only if (a, b, c, d, e, f) are tetrahedral.

In other words, the satisfaction of Eq. (2) for all Young diagram (α, β, δ) is actually *sufficient* to conclude (a, b, c, d, e, f) are tetrahedral. Along the way, we derive an analogous set of inequalities for the original Horn problem along with an accompanying distance bound. thm:horn_orig_poly_ineqs

Theorem 3. *Let (a, b, c) be a triple of non-negative eigenvalues. If there exists positive semidefinite matrices $A, B, C \geq 0$ with eigenvalues (a, b, c) satisfying $A + B = C$, then for all Young diagrams γ , we have*

$$\varphi_\gamma(c) \leq \sum_{\substack{\alpha, \beta \\ c_{\alpha\beta}^\gamma > 0}} \varphi_\alpha(a) \varphi_\beta(b). \quad (4) \quad \text{[eq:polynomial_horn_ineq]}$$

Moreover, if (a, b, c) satisfies both the trace condition Eq. (4) for all Young diagrams γ of size $|\gamma| = k$, then

$$\frac{1}{\text{Tr}[c]} D(a, b, c) \leq O\left(n\sqrt{\frac{\log k}{k}}\right). \quad (5)$$

where $D(a, b, c) := \min_{(A, B) \in \mathcal{O}_a \times \mathcal{O}_b} \|\text{eig}(A + B) - c\|_1$.

Eigenvalue estimation: Our approach to proving the above theorems relies on the concept of *weak Schur sampling*, a probabilistic technique developed in the context of quantum information theory for estimating the eigenvalues of density matrices (trace one, positive semidefinite matrices), which model the state of quantum systems [KW01; Chr08; OW16; HM02; CHW07; Hay17]. These techniques rely heavily on the concept of Schur-Weyl duality which describes the decomposition of the k -th tensor power of the n -dimensional complex vector space $\mathbb{C}^n$, which supports representations of both the symmetric group S_k by permutation of tensor factors and the group of unitary matrices $U(n)$ (or more generally invertible complex matrices $GL(n, \mathbb{C})$), into irreducible invariant subspaces indexed by Young diagrams $\lambda = (\lambda_1, \dots, \lambda_n)$ with k boxes and at most n rows. Given a positive semidefinite matrix $X \geq 0$ with eigenvalues $x = (x_1, \dots, x_n)$ the spectrum estimation technique describes how the so-called Schur-Weyl distribution $p_X(\lambda) = \frac{\text{Tr}[P_\lambda X^{\otimes k}]}{\text{Tr}[X^{\otimes k}]}$ concentrates strongly around those Young diagrams λ which are proportional to eigenvalues x and that this concentration strengthens as the sizes of the Young diagrams, k , tend to infinity. In this paper, we exploit this concentration phenomena to prove the distance bounds presented above.

6j-symbols Given a complex reductive group G with unitary irreducible representations (irreps) V_λ labelled by highest weights λ along with tensor-product multiplicity spaces, $M_{\alpha, \beta}^\gamma := \text{Hom}_G(V_\gamma, V_\alpha \otimes V_\beta)$, the 6j-symbols are maps between these multiplicity spaces which encode the associativity constraint $(V_\alpha \otimes V_\beta) \otimes V_\gamma \cong V_\alpha \otimes (V_\beta \otimes V_\gamma)$ between a triple of irreps, V_α, V_β , and V_δ of G . Specifically, a 6j-symbol depends on six irreps, $V_\alpha, V_\beta, V_\gamma, V_\delta, V_\epsilon$, and V_ϕ , and is the linear map between multiplicity spaces of the type

$$\left\{ \begin{matrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{matrix} \right\} : M_{\alpha, \beta}^\gamma \otimes M_{\delta, \gamma}^\epsilon \rightarrow M_{\alpha, \phi}^\epsilon \otimes M_{\beta, \delta}^\phi,$$

corresponding to the block-wise components of the canonical isomorphism from $\bigotimes_\gamma M_{\alpha, \beta}^\gamma \otimes M_{\delta, \gamma}^\epsilon$ to $\bigotimes_\phi M_{\alpha, \beta}^\gamma \otimes M_{\delta, \gamma}^\epsilon$. Whenever the group G is multiplicity-free, in the sense that the dimensions of all multiplicity spaces, $m_{\alpha, \beta}^\gamma := \dim M_{\alpha, \beta}^\gamma$, are either zero or one, the corresponding 6j-symbol is merely a scalar quantity.

Wigner's 6j-symbol for SU(2): The group SU(2) is multiplicity-free and the 6j-symbols for SU(2) are, depending on the normalization and/or phase convention, are also referred to as Wigner 6j-symbols or Wigner 6j-coefficients [Wig93] or the Racah W -coefficients [Rac42]. The 6j-symbols for SU(2) play a foundational role in the quantum theory of angular momentum, modelling the recoupling theory of three angular momenta, and thus they are used frequently in calculations involving the scattering of multiple particles in nuclear, atomic and molecular physics [VMK88; Edm96; Aqu+12]. Beyond the direct application to the theory of angular momenta, the 6j-symbols for SU(2) also constitute a primitive object in the theory of spin networks [Pen71; BS10], representing a minimal non-trivial closed spin network. General spin networks are networks of trivalent vertices and connections labelled by irreps (spins) of SU(2) which provide a discrete/combinatorial and manifestly invariant model with applications to gauge theories on graphs or manifolds [Bae96], invariants of knots and manifolds [Res87; TV92], topological quantum field theory [DJN93], quantum computing [MR05], quantum change-point discrimination [AH11], and various aspects of proposed models for quantum gravity [Bae99; BC98; Bar03; Rov04].

Throughout this paper, by the *asymptotics* of 6j-symbols, we are referring to their behaviour in the regime wherein the six highest weights, $\lambda \in \{\alpha, \beta, \gamma, \delta, \epsilon, \phi\}$, are becoming larger and larger in a suitable sense. The asymptotics of the 6j-symbol for SU(2), wherein the irreps of dimension $\ell + 1$ may be identified either by integers $\ell \in \{0, 1, \dots\}$ or half-integers $j = \frac{\ell}{2} \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$, were initially investigated by Wigner [Wig60, Chapter 27]. In the context of the quantum theory of angular momenta, the limit of large ℓ , or equivalently large spins j , may also be interpreted as the limit wherein Planck's constant tends to zero, $\hbar \propto \frac{1}{k} \rightarrow 0$, which suppresses the effects of quantizing angular momentum; therefore, these asymptotic limits are also known as semiclassical limits. Given three particles with spins $\frac{a}{2}$, $\frac{b}{2}$ and $\frac{d}{2}$ and total spin $\frac{e}{2}$, the

6j-symbol, $\left\{ \begin{smallmatrix} a & b & c \\ d & e & f \end{smallmatrix} \right\}$, can be interpreted as the quantum mechanical amplitude to observe the first and second particles to have combined spin $\frac{c}{2}$ given the second and third have combined spin $\frac{f}{2}$ [Wig60; Rob99]. Wigner considered the probability for this event to occur in the asymptotic limit of large spins, i.e., the square of the 6j-symbol, and conjectured the estimate:

$$\left\{ \begin{smallmatrix} a & b & c \\ d & e & f \end{smallmatrix} \right\}^2 \approx \frac{1}{3\pi V}$$

where V is the volume of the Euclidean tetrahedron with side-lengths (a, b, c, d, e, f) . Building upon the work of Wigner, Ponzano and Regge conjectured a formula for the amplitude of the 6j-symbol itself in the limit of large spins [PR68]. Later, using techniques from geometric quantization theory, Roberts rigorously proved an asymptotic formula which is equivalent to the conjecture of Ponzano and Regge (up to minor differences) [Rob99]:

Theorem 4 (Roberts). *If (a, b, c, d, e, f) are integer side-lengths of a Euclidean tetrahedron with volume V and corresponding exterior dihedral angles $(\theta_a, \theta_b, \theta_c, \theta_d, \theta_e, \theta_f)$, then*

$$\left\{ \begin{smallmatrix} ka & kb & kc \\ kd & ke & kf \end{smallmatrix} \right\} \stackrel{k \rightarrow \infty}{\sim} \sqrt{\frac{2}{3\pi V k^3}} \cos \left(\sum_{\ell} (k\ell + 1) \frac{\theta_{\ell}}{2} + \frac{\pi}{4} \right),$$

where the sum is over the six sides ℓ . Otherwise, if there does not exist a Euclidean tetrahedron with side-lengths (a, b, c, d, e, f) , then $\left\{ \begin{smallmatrix} ka & kb & kc \\ kd & ke & kf \end{smallmatrix} \right\}$ decays to zero at an exponential rate as $k \rightarrow \infty$.

It is worth emphasizing that Chen, Ismail, and Simeonov [CIS99] obtained a rigorous proof of the Ponzano and Regge asymptotics (with minor correction due to [Wat99]) slightly before Chen, Ismail, and Simeonov using estimates from the theory of hypergeometric functions. Additional proofs of this asymptotic correspondence between 6j-symbols for $SU(2)$ and Euclidean tetrahedra have emerged in recent years including the elementary proof by Gurau which applies Stirling's approximation directly to Racah's single-sum formula [Gur08] as well as group integral approach to evaluate so-called simple spin networks as Feynman graphs [FK00; FL03]. See also the paper by Bartlett and Nzaganya for an accessible presentation of these kinds of calculations [BN23]. Analogous asymptotic formulas have also been obtained for the 6j symbol for the quantum group $U_q(\mathfrak{sl}_2)$ for a k -dependent parameter q [TW06]. The asymptotics calculated by Taylor and Woodward encode (i) the existence of a spherical tetrahedron on the unit sphere with prescribed edgelengths whenever $q = \exp(i\pi/r(k))$ is an $r(k)$ th root of unity or (ii) the existence of a hyperbolic tetrahedron with prescribed edgelengths when $q = \exp(\pi/r(k))$ is positive real (here $r(k) = k(r-2) + 2$ for $r > 2$ a fixed integer) [TW06].

More generally, that the asymptotics of representation-theoretic quantities (such as 6j-symbols) are controlled by the existence of some critical geometric object is explained by the theory of geometric quantization [Rob02]. Geometric quantization, broadly speaking, associates classical phase spaces modelled by symplectic manifolds to quantum phase spaces modelled by Hilbert spaces [Rob02; GS82; Kir04]. In particular, Kirillov's orbit principle [Kir04] proposes an association between (i) the coadjoint orbits of a compact Lie group G passing through an integral weight λ to (ii) the corresponding unitary irreps of G of highest weight λ . Furthermore, products of coadjoint orbits correspond to tensor products of irreps and moreover symplectic quotients of the resulting products of orbits are identified with invariant vectors in the resulting tensor products of irreps [Rob02; GS82]. For the aforementioned case of $SU(2)$, the coadjoint orbits can be identified the $SO(3)$ rotations of vectors in Euclidean $\mathbb{R}^3$ (i.e. spheres of fixed radius), and thus the asymptotics of representation-theoretic quantities for $SU(2)$ are controlled by the existence of corresponding geometric objects in Euclidean $\mathbb{R}^3$ with prescribed edge-lengths [Rob02].

6j-symbols for $U(n)$: One application of the techniques developed in this paper is to provide estimates for the asymptotics of the 6j symbols for the group $U(n)$ (or equivalently for $SU(n)$) wherein the polynomial irreps V_{λ}^n are labelled by highest-weights which can also be identified with Young diagrams λ . One of the primary difficulties in proving statements about 6j symbols for $U(n)$ for $n \geq 3$ is the presence of non-trivial multiplicity spaces, i.e. the dimensions of tensor-product multiplicity spaces for $U(n)$, also known as Littlewood-Richardson coefficients, can be greater than one. Because of these difficulties, over the years,

considerable attention has been placed on the study of $U(n)$ (or $U_q(\mathfrak{sl}_n)$) $6j$ symbols which are multiplicity-free in the sense that the dimensions of each of the multiplicity spaces is either one or zero (in the latter case the $6j$ symbol defined to be zero). For instance, tensor products involving symmetric representations, where $\lambda = (k)$, or conjugates of symmetric representations, $\lambda = (k^{(n-1)})^1$, are necessarily multiplicity free; Nawata, Pichai, and Zodinmawia conjectured closed form expressions for the $U_q(\mathfrak{sl}_n)$ $6j$ symbols when a, b, d and e are integers and either $(\alpha, \beta, \delta, \epsilon) = ((a), (b^{(n-1)}), (d), (e^{(n-1)}))$ and $a + d = b + e$ (Type I) or $(\alpha, \beta, \delta, \epsilon) = ((a), (b), (d^{(n-1)}), (e^{(n-1)}))$ and $a + b = d + e$ (Type II) as well as the cases which can be related to these by known symmetries of the corresponding $6j$ symbol [NPZ13]. Building upon this work, Alekseev, Morozov, and Sleptsov studied both the asymptotics of and symmetries in each of the Type I and Type II cases [AMS20]. More recently, Alcock-Zeilinger et al. have derived formulae for the family of $U(n)$ $6j$ symbols where at least one pair of non-adjacent edges, say (α, δ) correspond to a pair of fundamental/defining representations, i.e., $\alpha, \delta = (1)$ [Alc+23]. Shortly thereafter, Keppeler, Plätzer, and Sjødahl expanded upon these results to derive formulae for the family of $U(n)$ $6j$ symbols where either (i) at least one pair of non-adjacent edges consists of fundamental or adjoint representations, i.e., $\alpha, \delta \in \{(1), (2, 1^{(n-2)})\}$, or (ii) at least one vertex, say (α, β, ϕ) , consists entirely of fundamental or adjoint representations, i.e., $\alpha, \beta, \phi \in \{(1), (2, 1^{(n-2)})\}$ [KPS24]. As stated by Coquereaux, “apart from very few special cases (low dimension of irreps and no multiplicity), ... we are not aware of any published general result.” [Coq23].

TC: Some residual research-notes concerning the $6j$ symbols:

- The graphical notion, specifically the use of an additional edge for multiplicity spaces can has been used for a long time, as early as [AB68; Mou84].
- Aquilanti et al. provides an excellent introduction to subject of classical $6j$ symbols, their history, diagrammatics, presentations and asymptotics [Aqu+12]
- There is a series of papers involving Alexey Sleptsov, starting with a computation of quantum $6j$ symbols for generic dimension n but with symmetric representations in terms of certain hypergeometric polynomials [MMS17], then with a discussion of the eigenvalue hypothesis and its potential connection to generalized Regge symmetries [MS18], then with a calculation of all multiplicity free quantum $6j$ symbols for arbitrary dimension [AMS20], and then finally with the discovery of novel “tug-the-hook” symmetries for quantum $6j$ symbols for all n [LS23].

Our approach bypasses the difficulties presented by multiplicities by focusing only on the *norms* of the $6j$ symbol for $U(n)$. Following the procedure provided by geometric quantization, one expects the asymptotics of the $6j$ -symbol for $U(n)$ to be controlled by existence of some kind of tetrahedral object in space of coadjoint orbits for $U(n)$. In particular, the coadjoint orbits of $U(n)$ can be identified with the orbits of $n \times n$ Hermitian matrices, X , under the conjugation action, $X \mapsto U^{-1}XU$, by unitary $n \times n$ matrices $U \in U(n)$; in other words, coadjoint orbits are associated to sets of $n \times n$ Hermitian matrices sharing a common set of eigenvalues $x = (x_1 \geq \dots \geq x_n)$. Using our methods we are able to prove the following inverse polynomial lower bound. thm: asym_6j

Theorem 5. *If (a, b, c, d, e, f) are the eigenvalues of $n \times n$ of positive semidefinite matrices (A, B, C, D, E, F) satisfying $A + B = C$, $B + D = F$, $C + D = E$, and $A + F = E$, then there exists a sequence of Young diagrams $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$ converging in proportion to the eigenvalues (a, b, c, d, e, f) such that*

$$\left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\| \geq \frac{1}{\text{poly}(k)}. \quad (6)$$

Otherwise, if (a, b, c, d, e, f) are not the eigenvalues of any such matrices, then for any sequence of Young diagrams converging in proportion to (a, b, c, d, e, f) , we have exponential decay in the value of $\left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|$.

Before continuing, it is worth emphasizing the similarities and differences between our result and the asymptotic formula proven by Roberts above [Rob99]. First and foremost, our result goes beyond that of Roberts in the sense that it applies to the higher-rank Lie group $U(n)$ for $n \geq 3$. Moreover, our result, even when restricted to the case for $SU(2)$, goes beyond the statement proven by Roberts in the sense our result is applicable to *irrational* eigenvalues/edge-lengths, whereas the former is only applicable to *integer*, or with

¹Here $k^{(n-1)}$ is shorthand for a string $k, k, \dots, k$ consisting of $n - 1$ repetitions of k .

a minor re-scaling, *rational* eigenvalues/edge-lengths. Nevertheless, one drawback of our result is that it merely promises the *existence* of some sequence of Young diagrams, $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$, converging to the eigenvalues (a, b, c, d, e, f) , whereas the result of Roberts provides a concrete statement about the particular sequence of *scaled* Young diagrams $(k\alpha, k\beta, k\gamma, k\delta, k\epsilon, k\phi)$.

Another drawback of our result is that it merely provides a polynomial lower bound on the value of the $\left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|$ and does not provide an asymptotic formula which would serve as the a proper generalization of the Ponzano and Regge formula above [PR68]. The Ponzano and Regge formula for $SU(2)$ was traditionally obtained using the method of *stationary phases* which relies on both an integral formula for the $6j$ -symbol with exponential integrand along with a geometric understanding of the *isolated* critical points for the exponential phase [PR68; Gur08]. In the case of $SU(2)$ with generic edge-lengths, there is a formulation involving exactly *two* isolated critical points corresponding the $SO(3)$ -orbit $[\tau]$ of a Euclidean tetrahedron τ with given edge-lengths and (ii) the $SO(3)$ -orbit $[\tau']$ of the Euclidean tetrahedron τ' which is the mirror-image of the tetrahedron τ with the same fixed side lengths. Generalizing this type of criticality analysis to $U(n)$ for $n \geq 3$ appears to be a significant challenge, and even if an analysis was performed, one would still need to compute a corresponding Hessian for each critical point and evaluate the total contribution [Rob02]. Perhaps another approach, such as the approach adopted by Taylor and Woodward (see also [CIS99]) to prove asymptotics for the $6j$ -symbol for $U_q(\mathfrak{sl}_2)$ using recursion relations [TW06] (see also the formulas for the $6j$ -symbols for $\mathfrak{gl}_n$ in terms of generalized hypergeometric functions due to [Art24; Art23; Art24]).

Quantum marginals & the $6j$ -symbol for the symmetric group S_k : The techniques used in this paper to derive bounds on the $6j$ symbol for the unitary group $U(n)$ are closely related to the techniques used in an earlier paper by Christandl, Şahinoğlu, and Walter which derived analogous bounds on the $6j$ symbol for the symmetric group S_k [CŞW18]. In that paper, the authors considered the problem of determining whether there exists a density matrix ρ_{ABC} with partial traces $\rho_{AB}, \rho_{BC}, \rho_A, \rho_B$, and ρ_C with a given sextuple of eigenvalues [CŞW18]. Such problems arise naturally in the context of quantum information theory and are typically referred to as quantum marginals problems [CŞW18; Fra22]. In particular, they proved the $6j$ symbols for the symmetric group S_k (referred to as the recoupling coefficients) in the limit as $k \rightarrow \infty$ either decay exponentially or can be bounded from below by a polynomial factor depending on whether there exists a density matrix with the prescribed marginal spectra [CŞW18]. Moreover, they also demonstrated how the tetrahedral Horn problem may be viewed as a special case of the quantum marginals problem and, in a complementary manner, furthermore demonstrated how, by restricting to the subgroup $S_k \subset U(k)$ of permutation matrices and applying Nishiyama’s restriction theorem [Nis00], the $6j$ symbols for unitary group $U(n)$ may be viewed as a special case of the $6j$ symbols for the symmetric group S_k [CŞW18]. This latter result obtained by [CŞW18] is a generalizes how the Littlewood-Richardson coefficients can be cast as a special case of the Kronecker coefficients [Lit58; Mur55; IP23]. The approach taken by this paper differs from that of [CŞW18] in that we prove the correspondence between solutions to the tetrahedral Horn problem on one hand and the asymptotics of the $6j$ symbol for $U(n)$ on the other hand *directly*, rather than first pass through the quantum marginals problem or through the $6j$ symbols for the symmetric group. We note that this more direct approach was initiated, in the $n = 2$ case, by Backens [Bac10].

Structure of the paper: TODO: Descide on the structure of the paper? Finally, we prove an upper bound on general $6j$ symbols using the Cauchy-Schwarz inequality which, in the $n \rightarrow \infty$ asymptotic limit, gives rise to a novel proof of a known strongly subadditive inequality involving unnormalized entropies.

2 Tetrahedral eigenvalues

sec:tet_eig

2.1 Eigenvalue problems for matrix sums

Throughout this paper, we consider the relationship between $n \times n$ Hermitian matrices and their real-valued tuples of eigenvalues. To denote the set of $n \times n$ Hermitian matrices, we use

$$\mathcal{H}(n) := \{X \in M_n(\mathbb{C}) \mid X^\dagger = X\}, \quad (7)$$

and to denote the set of possible eigenvalues of $n \times n$ matrices, we use

$$\text{Eig}(n) := \{x = (x_1, \dots, x_n) \in \mathbb{R}^n \mid x_1 \geq \dots \geq x_n\}. \quad (8)$$

Let $\mathcal{H}_+(n)$ be the subset of positive semidefinite $n \times n$ Hermitian matrices and $\text{Eig}^+(n)$ the corresponding set of non-negative eigenvalues. Given a Hermitian matrix $X \in \mathcal{H}(n)$, $\text{eig}(X)$ denotes the eigenvalues of X . The set of Hermitian matrices with eigenvalues $x \in \text{Eig}(n)$ is denoted by

$$\mathcal{O}_x := \{X \in \mathcal{H}(n) \mid \text{eig}(X) = x\}, \quad (9)$$

and $[x]$ denotes the diagonal matrix with entries $(x_1, \dots, x_n)$ along the diagonal:

$$[x] := \text{diag}(x_1, \dots, x_n). \quad (10)$$

The set of Hermitian matrices with eigenvalues x can then be identified with the unitary orbit of the diagonal matrix $[x]$, i.e., $\mathcal{O}_x = \{U[x]U^{-1} \mid U \in \text{U}(n)\}$. Moreover, for any function of Hermitian matrices, f , which unitarily invariant in the sense that $f(X) = f(UXU^{-1})$ holds for all $X \in \mathcal{H}(n)$ and $U \in \text{U}(n)$, we let $f[x]$ be shorthand for $f([x])$. For example,

$$\text{Tr}[x] := \text{Tr}([x]), \quad \text{and} \quad \text{rank}[x] := \text{rank}([x]). \quad (11)$$

prob:tri_problem

Definition 6 (Trigonal Horn problem). A triple $(a, b, c) \in \text{Eig}(n)^3$ of candidate eigenvalues are said to be *trigonal* if there exists a corresponding triple $(A, B, C) \in \mathcal{H}(n)^3$ of $n \times n$ Hermitian matrices satisfying (i) the linear constraint

$$A + B = C, \quad (12)$$

{eq:tri_linear_constraints}

and (ii) the three eigenvalue constraints

$$\text{eig}(A) = a, \quad \text{eig}(B) = b, \quad \text{eig}(C) = c. \quad (13)$$

{eq:tri_eigenvalue_constraints}

The problem of determining the set of trigonal eigenvalues (a, b, c) is known as *Horn's problem*.

prob:tet_problem

Definition 7 (Tetrahedral Horn problem). A sextuple $(a, b, c, d, e, f) \in \text{Eig}(n)^6$ of candidate eigenvalues are said to be *tetrahedral* if there exists a corresponding sextuple $(A, B, C, D, E, F) \in \mathcal{H}(n)^6$ of $n \times n$ Hermitian matrices satisfying (i) the four linear constraints

$$A + B = C, \quad B + D = F, \quad C + D = E, \quad A + F = E, \quad (14)$$

{eq:tet_linear_constraints}

and (ii) the six eigenvalue constraints

$$\text{eig}(A) = a, \quad \text{eig}(B) = b, \quad \text{eig}(C) = c, \quad \text{eig}(D) = d, \quad \text{eig}(E) = e, \quad \text{eig}(F) = f. \quad (15)$$

{eq:tet_eigenvalue_constraints}

The problem of determining the set of tetrahedral eigenvalues (a, b, c, d, e, f) will be referred to as the *tetrahedral Horn problem*.

The tetrahedral Horn problem can be viewed as a generalization of the Horn's problem in a variety of ways. For instance, a triplet (a, b, c) of eigenvalues is trigonal if and only if the sextuple $(a, b, c, 0, c, b)$ is tetrahedral. More generally, it is possible to view the tetrahedral Horn problem as the conjunction of four

separate instances of Horn's original problem subject to additional matching constraints. Specifically, a sextuple of eigenvalues (a, b, c, d, e, f) is tetrahedral only if each of the following four triplets are trigonal:

$$(a, b, c), \quad (b, d, f), \quad (c, d, e), \quad (a, f, e). \quad (\text{eq:four_faces}) \quad (16)$$

If each of the above triplets are trigonal, then there exists a set of twelve Hermitian matrices with the corresponding eigenvalues satisfying

$$A + B = C, \quad \tilde{B} + D = F, \quad \tilde{C} + \tilde{D} = E, \quad \tilde{A} + \tilde{F} = \tilde{E}. \quad (\text{eq:four_faces_matrices}) \quad (17)$$

The tetrahedral Horn problem goes beyond four separate instances of Horn's original problem in that it additionally requires the primed and unprimed Hermitian matrices to coincide:²

$$A = \tilde{A}, \quad B = \tilde{B}, \quad C = \tilde{C}, \quad D = \tilde{D}, \quad E = \tilde{E}, \quad F = \tilde{F}. \quad (18)$$

2.2 Deviation Measures

In order to quantify the degree to which a candidate collection of eigenvalues is compatible, either in the case of Horn's original problem or in the case of the tetrahedral Horn problem, it will be helpful to define some measures of deviation or distance which are non-negative and vanish if and only if the eigenvalues are compatible.

Proposition 8. *Let $1 \leq p \leq \infty$. Then for pairs of Hermitian matrices $X, Y \in \mathcal{H}(n)$ with eigenvalues $\text{eig}(X) = x = (x_1 \geq \dots \geq x_n)$ and $\text{eig}(Y) = y = (y_1 \geq \dots \geq y_n)$, one has*

$$\|X - Y\|_p \geq \|x - y\|_p. \quad (19)$$

Moreover, if $S \subseteq \mathcal{H}(n)$ is a unitarily invariant and closed subset of Hermitian matrices. Then

$$\min_{Y \in S} \|X - Y\|_p = \min_{y \in \text{eig}(S)} \|x - y\|_p, \quad (20)$$

In terms of Horn's original problem, we can consider the set S of Hermitian matrices which are of the form $A + B$ for some $(A, B) \in \mathcal{O}_a \times \mathcal{O}_b$.

Proposition 9. *For a given triple of eigenvalues, (a, b, c) , and $1 \leq p \leq \infty$, let*

$$D_p(a, b, c) := \min_{U, V \in \text{U}(n)} \|\text{eig}(U \cdot [a] + V \cdot [b]) - c\|_p. \quad (21)$$

Then $D_p(a, b, c) \geq 0$ with equality holding if and only if (a, b, c) are trigonal. Moreover,

$$D_p(a, b, c) = \min_{A, B, C \in \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_c} \|A + B - C\|_p. \quad (22)$$

For the tetrahedral Horn problem, there is some additional freedom in how one incorporates the various constraints. The measure of deviation which emerges naturally from our analysis is the following.

Definition 10. For a given sextuple of eigenvalues, (a, b, c, d, e, f) , and $1 \leq p \leq \infty$, let

$$D \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} := \min_{U, V, W \in \text{U}(n)} \left(\|\text{eig}(U \cdot [a] + V \cdot [b]) - c\|_1^2 + \|\text{eig}(V \cdot [b] + W \cdot [d]) - f\|_1^2 + \|\text{eig}(U \cdot [a] + V \cdot [b] + W \cdot [d]) - e\|_1^2 \right)^{\frac{1}{2}}. \quad (23)$$

Then $D \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \geq 0$ with equality holding if and only if (a, b, c, d, e, f) are trigonal.

²Note that by conjugating the third and fourth conditions in Eq. (17) by unitaries U and V respectively, it is possible to set $\tilde{C} \mapsto U^{-1}\tilde{C}U = C$ and $\tilde{F} \mapsto V^{-1}\tilde{F}V = F$, and thus one could assume without loss of generality that $C = \tilde{C}$ and $F = \tilde{F}$.

2.3 Low dimensional solutions

Remark 11 (Traces are tetrahedral). In the original version of Horn's problem, the eigenvalues (a, b, c) of Hermitian matrices (A, B, C) satisfying $A + B = C$ necessarily must satisfy a basic equality condition known as the *trace condition*:

$$\text{Tr}[a] + \text{Tr}[b] = \text{Tr}[c], \quad \text{\texttt{\{eq:tri_trace_cond\}}} \quad (24)$$

where $\text{Tr}[x] := \sum_i x_i$ is shorthand for the sum of the eigenvalues $x = (x_1 \geq \dots \geq x_n)$. This condition must hold because $A + B = C$ implies $\text{Tr}(C) = \text{Tr}(A + B) = \text{Tr}(A) + \text{Tr}(B)$ and $\text{Tr}(X) = \text{Tr}[x]$ for any Hermitian matrix X with eigenvalues x . For the tetrahedral Horn's problem, a set of tetrahedral eigenvalues (a, b, c, d, e, f) necessarily satisfy four *trace conditions*:

$$\text{Tr}[a] + \text{Tr}[b] = \text{Tr}[c], \quad \text{Tr}[b] + \text{Tr}[d] = \text{Tr}[f], \quad \text{Tr}[d] + \text{Tr}[c] = \text{Tr}[e], \quad \text{Tr}[a] + \text{Tr}[f] = \text{Tr}[e]. \quad \text{\texttt{\{eq:tet_trace_conds\}}} \quad (25)$$

Furthermore, for the trivial case where $n = 1$, the above trace constraints are both necessary and sufficient for the tetrahedral Horn problem. \texttt{exam:d_2}

Example 12 (Tetrahedral Horn's problem for $d = 2$). Here we articulate the solution to the tetrahedral Horn's problem for 2×2 Hermitian matrices. We restrict our focus to the case of *traceless* 2×2 Hermitian matrices³. In this context, the sextuple of eigenvalues (a, b, c, d, e, f) can be identified with a sextuple of non-negative real numbers $(a, b, c, d, e, f) \in \mathbb{R}_{\geq 0}^6$ such that the sextuple of traceless eigenvalues are

$$(a, -a), \quad (b, -b), \quad (c, -c), \quad (d, -d), \quad (e, -e), \quad (f, -f). \quad \text{\texttt{\{eq:traceless_two_dim_tetra\}}} \quad (26)$$

Moreover, each traceless 2×2 Hermitian matrix, X , may be identified with a three-dimensional vector $(x_1, x_2, x_3) \in \mathbb{R}^3$ using the standard Pauli-matrix basis $\{\sigma_1, \sigma_2, \sigma_3\}$ according to the formula:

$$X = x_1\sigma_1 + x_2\sigma_2 + x_3\sigma_3 = \begin{pmatrix} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{pmatrix}. \quad (27)$$

Under this identification, the eigenvalues of X are determined by the Euclidean length of the vector (x_1, x_2, x_3) :

$$\text{eig}(X) = \text{eig} \begin{pmatrix} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{pmatrix} = (\|x\|, -\|x\|), \quad \text{where,} \quad \|x\| = \sqrt{x_1^2 + x_2^2 + x_3^2}. \quad (28)$$

Furthermore, if A is determined by the coordinates (a_1, a_2, a_3) and B by the coordinates (b_1, b_2, b_3) , then their sum $C = A + B$ is determined by the coordinates $(c_1, c_2, c_3) = (a_1 + b_1, a_2 + b_2, a_3 + b_3)$. Consequently, in this setting, the tetrahedral Horn's problem reduces to problem of describing the space of all possible edge-lengths (a, b, c, d, e, f) of a Euclidean tetrahedron in $\mathbb{R}^3$. This problem admits of a well-studied solution [Ric02; WD09], namely, the conjunction of a quadruplet of triangle-inequalities

$$|a - b| \leq c \leq a + b, \quad |b - d| \leq f \leq b + d, \quad |d - c| \leq e \leq d + c, \quad |a - f| \leq e \leq a + f, \quad \text{\texttt{\{eq:quadruplet_triangle_constraints\}}} \quad (29)$$

(which ensure the possibility of forming a Euclidean triangle for each of the four faces), together with a single non-linear inequality of the form

$$\text{CM}(a, b, c, d, e, f) = \frac{1}{288} \det \begin{pmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & a^2 & c^2 & e^2 \\ 1 & a^2 & 0 & b^2 & f^2 \\ 1 & c^2 & b^2 & 0 & d^2 \\ 1 & e^2 & f^2 & d^2 & 0 \end{pmatrix} \geq 0. \quad \text{\texttt{\{eq:volume_positive_constraint\}}} \quad (30)$$

The quantity $\text{CM}(a, b, c, d, e, f)$ is known as a Cayley-Menger determinant and, when $\text{CM}(a, b, c, d, e, f)$ is positive and when the triangle inequalities above are also satisfied, $\text{CM}(a, b, c, d, e, f)$ is always equal to *square* of the volume of the corresponding tetrahedron [SJS18, Equation 5.2]⁴. Figure 2 depicts a three-dimensional projection of the solution space described by Eq. (29) and Eq. (30).

³In Section 2.4, Corollary 18, we will see that this restriction can be made without loss of generality.

⁴The triangle inequalities $|a - b| \leq c \leq a + b$ are also equivalent to the non-negativity of the *area*, $\text{CM}(a, b, c)$, of the triangle

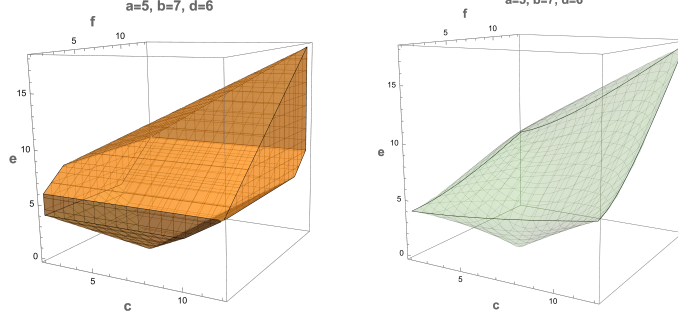


Figure 2: A slice of the set of admissible side-lengths (a, b, c, d, e, f) for Euclidean tetrahedron along the coordinates $(a, b, d) = (5, 7, 6)$ (Left) The polyhedron of values (c, e, f) satisfying the triangle inequalities for each face Eq. (29). (Right) The set of all values (c, e, f) satisfying both Eq. (29) and Eq. (30). tetrahedron(30)li_space_example

2.4 Symmetries of tetrahedral eigenvalues

sec:symmetries

There are numerous types of symmetries present in the statement of Definition 7. The purpose of this section is to describe the scaling, inversion, tetrahedral, and shifting symmetries of the problem. These symmetries can, in many cases, be used to simplify the problem by selecting a canonical representative of the corresponding equivalence class; in other cases, it will be useful to leave Definition 7 as stated. prop:scaling

Proposition 13 (Scaling symmetry). *Let (a, b, c, d, e, f) be a sextuple of candidate eigenvalues and let $s \in (0, \infty)$ be a positive real number. Then the scaled eigenvalues, (sa, sb, sc, sd, se, sf) , are tetrahedral if and only if the original, unscaled eigenvalues (a, b, c, d, e, f) are tetrahedral.*

Proof. The statement follows from the fact that $\text{eig}(sX) = s \text{eig}(X)$ holds for all Hermitian matrices X and positive scalars s and that each of the constraints in Eq. (14) are homogeneous in the six Hermitian matrices (A, B, C, D, E, F) ; for example, we have $sA + sB - sC = 0$ if and only if $s(A + B - C) = 0$ if and only if $A + B - C = 0$. □

One may wonder whether it is possible to extend the scaling symmetry to consider scaling by negative $s < 0$. Indeed, this can be accomplished, provided one accounts for the fact that negating a Hermitian matrices both negates and reverses the order of its eigenvalues. prop:inversion

Proposition 14 (Inversion symmetry). *Let (a, b, c, d, e, f) be a sextuple of candidate eigenvalues. Given any list of non-increasing eigenvalues $x = (x_1 \geq \dots \geq x_n)$, let x^* denote the inverted list of resorted and negated eigenvalues: $x^* = (-x_d \geq \dots \geq -x_1)$. Then (a, b, c, d, e, f) are tetrahedral if and only if $(a^*, b^*, c^*, d^*, e^*, f^*)$ are tetrahedral.*

Proof. The statement holds because (i) (A, B, C, D, E, F) satisfy Eq. (14) if and only if $(A', B', C', D', E', F') := (-A, -B, -C, -D, -E, -F)$ satisfy Eq. (14), and (ii) $\text{eig}(X) = x$ if and only if $\text{eig}(-X) = x^*$. □

The inversion symmetry Proposition 14 is just one element belonging to a group of 48 symmetries consisting of inversions of eigenvalues, $x \rightarrow x^*$, or permutations of elements in the sextuple (a, b, c, d, e, f) . prop:tet-symmetry

Proposition 15 (Oriented Tetrahedral symmetry). *The sextuple (a, b, c, d, e, f) of eigenvalues are tetrahedral if and only if any (and thus all) of the following sextuples of eigenvalues are tetrahedral:*

$$(a \ f \ e \ d^* \ c \ b) \quad (c \ e^* \ d^* \ f \ b \ a^*) \quad (31)$$

{eq:gens_of_tet_sym}

with side-lengths a, b and c where $\text{CM}(a, b, c)^2 = -\frac{1}{16} \det \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & a^2 & c^2 \\ 1 & a^2 & 0 & b^2 \\ 1 & c^2 & b^2 & 0 \end{pmatrix}$ is equivalent to Heron's formula.

Moreover, these two operations generate a finite group of order 48 whose action, grouped by conjugacy classes, produces:

$$\begin{array}{cccc}
(a & b & c & d & e & f) & (a^* & c & b & d & f & e) & (a & c^* & b^* & d^* & f^* & e^*) & (a^* & b^* & c^* & d^* & e^* & f^*) \\
(d & b & f & a & e & c) & (b^* & a^* & c^* & e & d & f) & (b & a & c & e^* & d^* & f^*) & (d^* & b^* & f^* & a^* & e^* & c^*) \\
(d^* & e & c & a^* & b & f) & (f^* & b & d^* & c^* & e^* & a^*) & (f & b^* & d & c & e & a) & (d & e^* & c^* & a & b^* & f^*) \\
(a & e^* & f^* & d & b^* & c^*) & (e & d^* & c & b^* & a & f^*) & (e^* & d & c^* & b & a^* & f) & (a^* & e & f & d^* & b & c) \\
(b & f^* & d^* & e & c & a) & (c & b^* & a & f & e & d) & (c^* & b & a^* & f^* & e^* & d^*) & (b^* & f & d & e^* & c^* & a^*) \\
(c & a^* & b & f^* & d^* & e^*) & (a & f & e & d^* & c & b) & (a^* & f^* & e^* & d & c^* & b^*) & (c^* & a & b^* & f & d & e) \\
(e & c^* & d & b & f & a^*) & (b & d & f & e^* & a^* & c^*) & (b^* & d^* & f^* & e & a & c) & (e^* & c & d^* & b^* & f^* & a) \\
(f & a & e & c^* & d & b^*) & (d & f^* & b^* & a^* & c^* & e^*) & (d^* & f & b & a & c & e) & (f^* & a^* & e^* & c & d^* & b) \\
(f^* & d & b^* & c & a & e) & (f & e^* & a^* & c & b & d^*) & (f^* & e & a & c^* & b^* & d) & (f & d^* & b & c^* & a^* & e^*) \\
(e^* & f & a^* & b^* & c^* & d) & (e^* & a & f^* & b & d^* & c) & (e & a^* & f & b^* & d & c^*) & (e & f^* & a & b & c & d^*) \\
(c^* & d^* & e^* & f & a^* & b) & (d^* & c^* & e^* & a & f^* & b^*) & (d & c & e & a^* & f & b) & (c & d & e & f^* & a & b^*) \\
(b^* & c & a & e^* & f^* & d^*) & (c^* & e & d & f^* & b^* & a) & (c & e^* & d^* & f & b & a^*) & (b & c^* & a^* & e & f & d)
\end{array} \tag{32}$$

Proof. To begin, notice that each of the operations on (a, b, c, d, e, f) appearing in Eq. (31) are invertible (indeed the first operation has order two, while the second operation has order four). The claim that these two operations generate a finite group of order 48 can be checked explicitly using computer software. Now suppose (a, b, c, d, e, f) are the eigenvalues of a sextuple of Hermitian matrices (A, B, C, D, E, F) satisfying Eq. (14). Then, for example, the sextuple $(A', B', C', D', E', F') = (A, F, E, -D, C, B)$ (which has the eigenvalues (a, f, e, d^*, c, b)) is also tetrahedral because

$$\begin{aligned}
A' + B' - C' &= (A + F - E) = 0, \\
B' + D' - F' &= -(B + D - F) = 0, \\
C' + D' - E' &= -(C + D - E) = 0, \\
A' + F' - E' &= (A + B - C) = 0.
\end{aligned} \tag{33}$$

Moreover, as the operation mapping (a, b, c, d, e, f) to (a^*, c, b, d, f, e) is invertible, one also has that if (a, b, c, d, e, f) is not tetrahedral, then neither is (a^*, c, b, d, f, e) . An analogous argument holds for the operation mapping (a, b, c, d, e, f) to (c, e^*, d^*, f, b, a^*) with (A', B', C', D', E', F') given by $(C, -E, -D, F, B, -A)$. $\square$

Remark 16. The group G of order 48 appearing in Proposition 15 corresponds to the subgroup of all possible signed permutations of 6 elements generated by permutations and negations of the elements of the sextuple (A, B, C, D, E, F) which preserves the set of linear constraints in Eq. (14). The group G is isomorphic to both the direct product $S_2 \times S_4$ and to the signed permutation group of three elements, i.e., $(S_2 \times S_2 \times S_2) \rtimes S_3$.

The first isomorphism, $G \simeq S_2 \times S_4$, can be understood as S_2 acting by the inversion of all eigenvalues,

$$(a, b, c, d, e, f) \leftrightarrow (a^*, b^*, c^*, d^*, e^*, f^*), \tag{34}$$

while S_4 acts by permutation of the *vertices* of the tetrahedron in Fig. 1: each vertex permutation induces a permutation of the six directed edges with the possibility of inverting the orientation of a subset of the edges (such edge orientation reversals can be accounted for a corresponding inversion of eigenvalues).

The second isomorphism, $G \simeq (S_2 \times S_2 \times S_2) \rtimes S_3$, can be understood by considering the triplet $(A, D), (B, E), (C, F)$ of non-adjacent pairs of edges in Fig. 1. Accordingly, it is helpful to arrange the sextuple (a, b, c, d, e, f) into a 2×3 grid using the following identification:

$$(a \ b \ c \ d \ e \ f) \cong \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}. \tag{35}$$

Then one can see that each S_2 factor in the direct product $S_2 \times S_2 \times S_2$ acts faithfully on the sextuple by both negating a certain subset of the columns and permuting the upper and lower rows leaving just one column unchanged.

$$\begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \leftrightarrow \begin{pmatrix} a & e^* & f^* \\ d & b^* & c^* \end{pmatrix}, \quad \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \leftrightarrow \begin{pmatrix} d & b & f \\ a & e & c \end{pmatrix}, \quad \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \leftrightarrow \begin{pmatrix} d^* & e & c \\ a^* & b & f \end{pmatrix}. \tag{36}$$

The symmetric group S_3 of order $3! = 6$ acts instead by permuting the columns and negating a subset of the entries in the top row, producing a subgroup of order 6 of the form:

$$\begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \quad \begin{pmatrix} b & c^* & a^* \\ e & f & d \end{pmatrix} \quad \begin{pmatrix} c^* & a & b^* \\ f & d & e \end{pmatrix} \\ \begin{pmatrix} a^* & c & b \\ d & f & e \end{pmatrix} \quad \begin{pmatrix} b^* & a^* & c^* \\ e & d & f \end{pmatrix} \quad \begin{pmatrix} c & b^* & a \\ f & e & d \end{pmatrix} \quad (37)$$

We further remark that these symmetries also corresponding to symmetries of the Wigner $6j$ operator $\begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix}$ in Definition 41 with the involution $\lambda \mapsto \lambda^*$ corresponding taking the dual representation $V_\lambda^n \mapsto (V_\lambda^n)^*$. In the case where $n = 2$, the involution map becomes the identity map (as every irreducible representation of $SU(2)$ is self-dual, or the reversal of a pair $(x, -x)$ of traceless eigenvalues is trivial), and thus the symmetry group collapses to a group of order $|S_4| = 4! = 24$. prop:shifting

Proposition 17 (Shifting symmetry). *Let (a, b, c, d, e, f) be a sextuple of candidate eigenvalues and let $x, y, z \in \mathbb{R}$ be fixed. Given any list of non-increasing eigenvalues $x = (x_1 \geq \dots \geq x_n)$, let $x + x := (x_1 + x \geq \dots \geq x_d + x)$. Then define shifted eigenvalues by the formula*

$$\begin{aligned} \tilde{a} &:= a + x, & \tilde{b} &:= b + y, & \tilde{c} &:= c + x + y, & \text{\textcolor{blue}{eq:shifted_tet_eigenvalues}} \\ \tilde{d} &:= d + z, & \tilde{e} &:= e + x + y + z, & \tilde{f} &:= f + y + z. \end{aligned} \quad (38)$$

The shifted eigenvalues $(\tilde{a}, \tilde{b}, \tilde{c}, \tilde{d}, \tilde{e}, \tilde{f})$ are tetrahedral if and only if the unshifted eigenvalues (a, b, c, d, e, f) are tetrahedral.

Proof. Given a Hermitian matrix X with eigenvalues $\text{eig}(X) = x$ and a constant $w \in \mathbb{R}$, the eigenvalues of $X + wI$ (where I is the identity matrix) are

$$\text{eig}(X + wI) = x + w = (x_1 + w \geq \dots \geq x_d + w). \quad (39)$$

The statement then follows because

$$(A + xI, B + yI, C + (x + y)I, D + zI, E + (x + y + z)I, F + (y + z)I) \quad (40)$$

satisfy the tetrahedral constraints in Eq. (14) if and only if (A, B, C, D, E, F) satisfy the tetrahedral constraints in Eq. (14) $\square$

As a corollary of the shifting symmetries (Proposition 17) one concludes that to solve the tetrahedral Horn's problem, in a certain sense, it suffices to solve the tetrahedral Horn's problem merely for traceless matrices (Corollary 18) or, alternatively, for positive definite or positive semidefinite matrices (Corollary 19). cor:traceless

Corollary 18 (Traceless eigenvalues). *Let (a, b, c, d, e, f) be a sextuple of candidate eigenvalues and for any list of n eigenvalues $x = (x_1 \geq \dots \geq x_n)$, let $x^0 := x - \text{Tr}[x]/n$ denote the traceless list of eigenvalues obtained by shifting x by $-\text{Tr}[x]/n$. Then (a, b, c, d, e, f) are tetrahedral if and only if (i) their traces $(\text{Tr}[a], \text{Tr}[b], \text{Tr}[c], \text{Tr}[d], \text{Tr}[e], \text{Tr}[f])$ are tetrahedral (they satisfy Eq. (25)) and (ii) $(a^0, b^0, c^0, d^0, e^0, f^0)$ are tetrahedral.*

Proof. This “only if” portion of the statement follows from both the tetrahedral trace conditions (Eq. (25)) and applications of the shifting symmetry Proposition 17 with (x, y, z) values

$$x = -n^{-1} \text{Tr}(A), \quad y = -n^{-1} \text{Tr}(B) \quad z = -n^{-1} \text{Tr}(D). \quad (41)$$

The “if” portion of the statement follows from the shifting symmetry applied to the negated (x, y, z) values. $\square$

The next corollary of the shifting symmetry proves that to solve the tetrahedral Horn's problem it suffices to solve the tetrahedral Horn's problem for positive definite spectra only. Indeed, one can simply take the shifting parameters (x, y, z) appearing in Proposition 17 to be sufficiently large. The following corollary

establishes, in a certain sense, the minimum shifting required to ensure strictly positive eigenvalues for all six Hermitian matrices. Note that in subsequent sections we will often explicitly assume the candidate eigenvalues (a, b, c, d, e, f) are either non-negative or positive for the sake of technical convenience or sometimes to establish a connection with density matrices from quantum theory (which are positive semidefinite matrices with unit trace).

cor:psd

Corollary 19 (Positive eigenvalues). *If (a, b, c, d, e, f) are a tetrahedral sextuple of eigenvalues, then necessarily the smallest eigenvalues satisfy*

$$c_d \geq a_d + b_d, \quad f_d \geq b_d + d_d, \quad e_d \geq d_d + c_d, \quad e_d \geq a_d + f_d. \quad \text{\textcolor{blue}{eq:min_eig_ineqs}} \quad (42)$$

Moreover, if (a, b, c, d, e, f) are a sextuple of candidate eigenvalues satisfying Eq. (42), then for any $(x, y, z) \in \mathbb{R}^3$ satisfying

$$x > -a_d, \quad y > -b_d, \quad z > -d_d, \quad \text{\textcolor{blue}{eq:positive_shift}} \quad (43)$$

the resulting shifted eigenvalues $(\tilde{a}, \tilde{b}, \tilde{c}, \tilde{d}, \tilde{e}, \tilde{f})$ (given by Eq. (38)) are all positive.

Proof. That tetrahedral eigenvalues satisfy Eq. (42) follows from four applications of the well-known inequality $c_d \geq a_d + b_d$ for Horn's problem where $C = A + B$ [Bha01]. As a reminder, the smallest eigenvalue x_n of a matrix X is characterized by a minimization: $x_d = \min_{\|v\|=1} \langle v, Xv \rangle$ which produces

$$c_d = \min_{\|v\|=1} \langle v, Cv \rangle = \min_{\|v\|=1} (\langle v, Av \rangle + \langle v, Bv \rangle) \geq \min_{\|v\|=1} \langle v, Av \rangle + \min_{\|w\|=1} \langle w, Bw \rangle = a_d + b_d. \quad (44)$$

Four applications of this inequality produces Eq. (42). Now if (x, y, z) satisfy Eq. (43), we have for all j

$$a_j + x \geq a_d + x > 0, \quad (45)$$

meaning $a + x = (a_1 + x \geq \dots \geq x_d + x)$ is strictly positive. Similar statements hold for b, c, d, e and f ; for instance, for all j ,

$$e_j + x + y + z \geq e_d + x + y + z > e_d - a_d - b_d - d_d \geq 0, \quad (46)$$

meaning $e + x + y + z$ is strictly positive also. $\square$

Remark 20. Although this section listed a handful of symmetries of the tetrahedral Horn's problem, there remains the possibility of additional symmetries which are unknown at this time. For example, in the case 2×2 Hermitian matrices, an additional class of symmetries on the set of edge-lengths (a, b, c, e, d, f) of tetrahedra, known as the *Regge symmetries*, hold [AI19; Boa07; PR68, Theorem 1.1]. If there exists a tetrahedron with edge lengths (a, b, c, d, e, f) , then there also exists a tetrahedron with edge lengths

$$(s_1 - a, s_1 - b, c, s_1 - d, s_1 - e, f) \quad \text{where} \quad s_1 = \frac{a + b + d + e}{2}. \quad (47)$$

Combining this symmetry with the permutational symmetries given by Proposition 15 (and noting that for the eigenvalues x of traceless 2×2 Hermitian matrices, $x = (x, -x)$, we have $x^* = x$) one concludes the co-existence of a family of six tetrahedra with edge lengths given by [VMK88, Chapter 9]:

$$\begin{aligned} (a, b, c, d, e, f) & \quad (s_1 - a, s_1 - b, c, s_1 - d, s_1 - e, f) \\ (a, s_3 - b, s_3 - c, d, s_3 - e, s_3 - f) & \quad (s_2 - a, b, s_2 - c, s_2 - d, e, s_2 - f) \\ (s_2 - d, s_1 - e, s_3 - f, s_2 - a, s_1 - b, s_3 - c) & \quad (s_1 - d, s_3 - e, s_2 - f, s_1 - a, s_3 - b, s_2 - c) \end{aligned} \quad (48)$$

where

$$s_1 = \frac{a + b + d + e}{2}, \quad s_2 = \frac{a + c + d + f}{2}, \quad s_3 = \frac{b + c + e + f}{2}. \quad (49)$$

Remark 21. The Regge symmetry of a tetrahedron can be understood as a consequence of an earlier discovery of a suprising symmetry of Wigner $6j$ symbol for $SU(2)$ originally due to Regge [Reg59]. The paper due to Boalch provides a comprehensive and modern treatment of the Regge symmetry for the $6j$ symbol for $SU(2)$ by viewing it as a consequence of a simpler symmetry of the $6j$ -symbol for $SU(3)$ [Boa07]. That the Regge symmetry of a tetrahedron emerges from the Regge symmetry of the associated $6j$ symbols

can be viewed as a consequence of the semiclassical limit of the latter [PR68; Rob99]. It has also been shown that the Regge symmetry of Euclidean tetrahedra is actually a scissors-congruence preserving both the volume and Dehn invariants [AI19].

At the moment, it remains unclear if there exists analogous symmetries for the generalized Wigner $6j$ symbol for $n \geq 3$ (although there are conjectured symmetries for the $6j$ symbols for $U_q(\mathfrak{sl}_n)$ which would follow from the eigenvalue hypothesis [MS18; LS23]). It could also be possible for additional symmetries to be present in the tetrahedral Horn problem for $n \geq 3$ which are *not* present in the $6j$ symbol for $U(n)$.

3 New inequalities for tetrahedral eigenvalues

The purpose of this section is to develop the technical machinery used to prove numerous new inequalities constraining the set of tetrahedral eigenvalues. Our approach relies heavily on the theory of irreducible representations of the group $U(n)$ of unitary $n \times n$ matrices, or equivalently, the theory of polynomial irreducible representations of the group $GL(n, \mathbb{C})$ of invertible $n \times n$ matrices with complex entries. More specifically, our approach relies on the decomposition of tensor powers $X^{\otimes k}$ of $n \times n$ matrices provided by Schur-Weyl duality. In Section 3.1 we review this prerequisite representation theory and provide elementary results regarding the decomposition of tensor powers $X^{\otimes k}$ and then afterwards, in Section 3.2, we describe how these decompositions of $X^{\otimes k}$ encode information about the eigenvalues x of X . Building upon this machinery, in Section 3.3, we state and prove a binomial theorem concerning various decompositions of tensors powers of matrix sums $(X + Y)^{\otimes k}$. Using the binomial theorem, in Section 3.4, we will derive new inequalities constraining the set of eigenvalue triples (α, β, γ) of positive semidefinite matrices (A, B, C) satisfying $A + B = C$ relevant for Horn's original eigenvalue problem. Afterwards, in Section 3.6, we again leverage the binomial theorem (or rather its multinomial generalization) to derive new inequalities constraining the set of eigenvalue sextuples (a, b, c, d, e, f) of positive semidefinite matrices (A, B, C, D, E, F) satisfying $A + B = C$, $B + D = F$, $C + D = E$, and $A + F = E$ relevant for the tetrahedral Horn problem.

3.1 Schur-Weyl duality

`sec:schur_weyl`

The purpose of this section is to review the representation theory of $U(n)$, or equivalently, the polynomial representation theory of $GL(n)$. The isomorphism classes of irreducible polynomial representations of $GL(n)$ are indexed by Young diagrams.

A Young diagram λ of size k is a collection of k boxes arranged in rows of sorted length and can be identified with its sequence of non-zero row-lengths, $\lambda \simeq (\lambda_1, \dots, \lambda_\ell)$, where λ_i counts the number of boxes in the i th row, $\lambda_1 \geq \dots \geq \lambda_\ell > 0$, and where $\sum_i \lambda_i = k$. The number of non-zero rows in λ is called the depth or length of λ and is denoted by $\ell(\lambda)$ and the size of λ is denoted by $|\lambda|$. As a shorthand, we write $\lambda \vdash_n k$ to denote a Young diagram λ of size $|\lambda| = k$ with depth at most $\ell(\lambda) \leq n$. For example, $\lambda = \mathbb{F}^3 \simeq (4, 2, 1)$ is a Young diagram of size $|\lambda| = 4 + 2 + 1 = 7$ and depth 3.

For each Young diagram ℓ of size k with depth at most n , there is an irreducible polynomial representation of $U(n)$ (or $GL(n, \mathbb{C})$), denoted by $(\pi_\lambda, V_\lambda^n)$, whose matrix coefficients are polynomials of homogeneous degree k in the entries of $U \in U(n)$. The dimension of the V_λ^n is given by Weyl's dimension formula for $U(n)$ [Hal08; IN66],

$$\dim(V_\lambda^n) = \frac{\Delta(\lambda + \rho)}{\Delta(\rho)} = \frac{\prod_{1 \leq i < j \leq n} (\lambda_i - i - \lambda_j + j)}{\prod_{1 \leq i < j \leq n} (j - i)}, \quad \text{eq:weyl_dim} \quad (50)$$

where $\rho \simeq (n-1, n-2, \dots, 1, 0)$ is known as the Weyl vector and where

$$\Delta(x_1, \dots, x_n) = \prod_{1 \leq i < j \leq n} (x_i - x_j). \quad (51)$$

is known as the Vandermonde determinant. The defining representation of $U(n)$, wherein $U \in U(n)$ acts via matrix-vector multiplication on $\mathbb{C}^n$, is associated to the Young diagram with one box, $\lambda = \square \simeq (1)$, and as dimension $\dim(V_\square^n) = n$ as $V_\square^n \simeq \mathbb{C}^n$.

The so-called Schur-Weyl decomposition of $(V_\square^n)^{\otimes k} \simeq (\mathbb{C}^n)^{\otimes k}$ into irreducible representations of $U(n)$ is as follows.

Proposition 22 (Schur-Weyl duality). *Let k and n be positive integers. Then the decomposition of $(\mathbb{C}^n)^{\otimes k}$ into irreducible subrepresentations of $S_k \times U(n)$, known as the Schur-Weyl decomposition, is given by*

$$(\mathbb{C}^n)^{\otimes k} \cong \bigoplus_{\lambda \vdash_n k} W_\lambda \otimes V_\lambda^n, \quad \text{eq:schur_weyl_decomp} \quad (52)$$

where the direct summands are indexed by Young diagrams λ of size k and depth at most n rows, and where W_λ supports an irreducible representation of S_k and V_λ^n supports the irreducible representation of $U(n)$.

The dimension W_λ can be computed using the Frobenius dimension formula [GW00, Corollaries 9.1.5],

$$\dim(W_\lambda) = \frac{|\lambda|! \Delta(\lambda + \rho)}{\prod_{i=1}^n (\lambda + \rho)_i!}. \quad \text{\texttt{\{eq:frob_dim\}}} \quad (53)$$

While this formula is useful for comparisons with $\dim(V_\lambda^n)$, it is not obviously independent of n for $n \geq \ell(\lambda)$. Indeed, the right hand side of Eq. (53) does not depend on n for $n \geq \ell(\lambda)$ and coincides with the Hook-length formula [GW00, Corollary 9.1.6], in this case

$$\dim(W_\lambda) = \frac{|\lambda|!}{H_\lambda}, \quad H_\lambda = \prod_{b \in \lambda} h_\lambda(b), \quad \text{\texttt{\{eq:hook_dim\}}} \quad (54)$$

where the hook length, $h_\lambda(b)$, of b counts the number of boxes to the right and below b plus one for itself. We refer to H_λ as the hook-product for λ .

Definition 23. Let $(\pi_\lambda, V_\lambda^n)$ be an irreducible representation of $U(n)$ labelled by the Young diagram λ with depth at most n . Let the $S_k \times U(n)$ equivariant isometry for the λ -isotypic subspace of $V_\square^{\otimes |\lambda|} \simeq (\mathbb{C}^n)^{\otimes |\lambda|}$ be

$$\iota_\lambda : W_\lambda \otimes V_\lambda^n \rightarrow V_\square^{\otimes |\lambda|}. \quad (55)$$

Then given an $n \times n$ matrix X , viewed as a linear operator on the defining representation $V_\square^n \simeq \mathbb{C}^n$, let $\pi_\lambda(X)$ be defined by

$$\pi_\lambda(X) := \frac{\text{Tr}_{W_\lambda}[\iota_\lambda^\dagger X^{\otimes |\lambda|} \iota_\lambda]}{\dim(W_\lambda)}.$$

Note that when X is invertible, i.e. $X \in \text{GL}(n, \mathbb{C})$, $\pi_\lambda(X)$ as defined above coincides with the representation of X on V_λ^n , so this abuse of notation is permitted. For any positive integer k , the k th tensor power of X then admits a decomposition as $X^{\otimes k} \simeq \bigoplus_{\lambda \vdash_n k} \pi_\lambda(X)^{\oplus W_\lambda}$, or more precisely,

$$X^{\otimes k} = \sum_{\lambda \vdash_n k} \iota_\lambda(\pi_\lambda(X) \otimes \text{id}_{W_\lambda}) \iota_\lambda^\dagger. \quad (56)$$

As a special case, when $\lambda = \square$, then $\pi_\square(X) = X$. Note also that the matrix coefficients of $\pi_\lambda(X)$ are homogeneous degree $|\lambda|$ polynomials in the matrix coefficients of X meaning for any scalar $s \in \mathbb{C}$,

$$\pi_\lambda(sX) = s^{|\lambda|} \pi_\lambda(X). \quad (57)$$

Additionally, if X, Y are two matrices, we have

$$\pi_\lambda(XY) = \pi_\lambda(X) \pi_\lambda(Y). \quad (58)$$

The corresponding character $\chi_\lambda(X) = \text{Tr}(\pi_\lambda(X))$ of a representation of a Hermitian matrix X with eigenvalues x is equal to

$$\chi_\lambda(X) := \text{Tr}(\pi_\lambda(X)) = \frac{\text{Tr}[P_\lambda X^{\otimes |\lambda|}]}{\dim(W_\lambda)} = s_\lambda(x),$$

where $P_\lambda := \iota_\lambda^\dagger \iota_\lambda$ is the orthogonal projection operator onto the λ -isotypic subspace, and where $s_\lambda(x) = s_\lambda(x_1, \dots, x_n)$ Schur-polynomial associated to λ .

There are various formulas for the Schur-polynomial $s_\lambda(x_1, \dots, x_n)$. Here we present the Jacobi's bialternant formula

$$s_\lambda(x) = \frac{\det[x_i^{\lambda_j + n - j}]}{\det[x_i^{n - j}]} = \frac{1}{\Delta(x)} \det[x_i^{(\lambda + \rho)_j}]. \quad (59)$$

3.2 Eigenvalue estimation & separation

sec:eig_est_and_sep

In this section we show how the representations $\pi_\lambda(X)$ of Hermitian matrices X encode useful information about the eigenvalues $x = \text{eig}(X)$ of X . These results apply to the case where $X \geq 0$ is a positive semidefinite matrix, in which case the representation $\pi_\lambda(X) \geq 0$ is positive semidefinite also. Consequently, one obtains a formula for each Hermitian matrix X with eigenvalues x and each positive integer k :

$$\text{Tr}(X)^k = \text{Tr}(X^{\otimes k}) = \sum_{\lambda \vdash k} \text{Tr}[P_\lambda X^{\otimes k}] = \sum_{\lambda \vdash k} \dim W_\lambda \text{Tr}[\pi_\lambda(X)] = \sum_{\lambda \vdash k} \dim W_\lambda s_\lambda(x). \quad (60)$$

Whenever $X \geq 0$, each summand above is also non-negative and thus one can define a probability distribution known as the *Schur-Weyl* distribution [OW16].

Definition 24. The Schur-Weyl distribution of order k for a positive semidefinite matrix $X \geq 0$ with eigenvalues $x = (x_1, \dots, x_n)$ is defined as the probability distribution over Young diagrams λ with k boxes with probability

$$p_X(\lambda) := \frac{\text{Tr}[P_\lambda X^{\otimes k}]}{\text{Tr}[X^{\otimes k}]} = \frac{\dim W_\lambda s_\lambda(x)}{\text{Tr}[x]^k} = \frac{|\lambda|!}{\text{Tr}[x]^k} \frac{s_\lambda(x)}{H_\lambda}. \quad (61)$$

A key property of the Schur-Weyl distribution $p_X(\lambda)$ is that it behaves similarly to the multinomial distribution $m_x(\lambda)$ where

$$m_x(\lambda) := \frac{|\lambda|!}{\text{Tr}[x]^k} \frac{x_1^{\lambda_1}}{\lambda_1!} \frac{x_2^{\lambda_2}}{\lambda_2!} \cdots \frac{x_n^{\lambda_n}}{\lambda_n!}. \quad (62)$$

prop:multinomial_bound

Proposition 25. Given a Young diagram λ and non-negative eigenvalues x ,

$$p_X(\lambda) \leq \dim V_\lambda^n m_{\underline{x}}(\lambda). \quad (63)$$

Proof. The proof follows from (i) an upper bound on the Schur-polynomial $s_\lambda(x)$,

$$s_\lambda(x) \leq \dim V_\lambda^n x_1^{\lambda_1} x_2^{\lambda_2} \cdots x_n^{\lambda_n}, \quad (64)$$

which can be expressed as a sum over $\dim V_\lambda^n$ multinomials of the form $x_1^{\mu_1} x_2^{\mu_2} \cdots x_n^{\mu_n}$ with $x_1^{\lambda_1} x_2^{\lambda_2} \cdots x_n^{\lambda_n}$ the largest, and (ii) a lower bound on the Hook-product,

$$H_\lambda \geq \lambda_1! \cdots \lambda_n!, \quad (65)$$

which follows because $h_\lambda(b) \geq \lambda_i - j + 1$ for a box in row i and column j . $\square$

Remark 26. One can eliminate the factor of $\dim V_\lambda^n$ using the dimension bound

$$\dim V_\lambda^n \leq (k+1)^{\frac{n(n-1)}{2}}. \quad (66)$$

The following known result is known as eigenvalue or spectrum estimation [ARS88; KW01; HM02].

Proposition 27 (Eigenvalue estimation). *Let $X \geq 0$ be a positive semidefinite $n \times n$ matrix with eigenvalues $x = (x_1 \geq \cdots \geq x_n \geq 0)$. Then the Schur-Weyl distribution $p_X(\lambda)$ for $\lambda \vdash k$ satisfies*

$$p_X(\lambda) \leq (k+1)^{\frac{n(n-1)}{2}} \exp(-kD(\underline{\lambda}||\underline{x})) \quad (67)$$

where $D(\underline{\lambda}||\underline{x})$ denotes the Kullback-Leibler divergence between the normalized Young diagram $\underline{\lambda} = \lambda/|\lambda|$ and the normalized eigenvalues $\underline{x} = x/\text{Tr}[x]$.

This result captures the fact that, like the multinomial distribution $m_x(\lambda)$, the Schur-Weyl distribution $p_X(\lambda)$ is sharply concentrated around Young diagrams λ whose distribution, $\underline{\lambda}$, is similar to the eigenvalue distribution $\underline{x}$.

For our purposes, it will be useful to use a related result which captures the fact if two positive semidefinite matrices $X, Y \geq 0$ have dissimilar eigenvalue distributions then the fidelity between their Schur-Weyl

distributions is exponentially small in k . Recall the fidelity $F(p, q) \in [0, 1]$ between distributions p, q , defined by

$$F(p, q) = \sum_i \sqrt{p_i q_i}, \quad (68)$$

is equal to $F(p, q) = 1$ if and only if $p = q$. We refer to this result throughout the paper as the eigenvalue separation lemma.

Lemma 28 (Eigenvalue separation). *Let (x, y) be eigenvalues of positive semidefinite $n \times n$ matrices $X, Y \geq 0$. Then the fidelity between Schur-Weyl distributions p_X and p_Y of order k*

$$F_k(p_X, p_Y) := \sum_{\lambda \vdash k} \sqrt{p_X(\lambda) p_Y(\lambda)} \quad (69)$$

is bounded by

$$F_k(p_X, p_Y) \leq (k+1)^{n(n-1)/2} F(\underline{x}, \underline{y})^k. \quad (70)$$

where $F(\underline{x}, \underline{y})$ is the fidelity between the eigenvalue distributions of X and Y ,

$$F(\underline{x}, \underline{y}) = \sum_{i=1}^n \sqrt{\frac{x_i}{\text{Tr}[X]} \frac{y_i}{\text{Tr}[Y]}}. \quad (71)$$

Proof. The proof follows from Proposition 25 and the fact that fidelities are multiplicative with respect to multinomial distributions, i.e. $F_k(m_x, m_y) \leq F(\underline{x}, \underline{y})^k$.

$$F_k(p_X, p_Y) \leq (k+1)^{n(n-1)/2} F_k(m_x, m_y) = (k+1)^{n(n-1)/2} F(\underline{x}, \underline{y})^k. \quad (72)$$

□

Remark 29. The eigenvalue separation lemma states that when two positive semidefinite matrices $X, Y \geq 0$ have the same trace $\tau = \text{Tr}(X) = \text{Tr}(Y)$ but distinct eigenvalues $x \neq y$, then we can conclude $F(\underline{x}, \underline{y}) < 1 - \varepsilon$ for some ε and therefore as $k \rightarrow \infty$, $F_k(p_X, p_Y) \rightarrow 0$.

Remark 30. In order to quantify the separation between distributions more geometrically, we use the fact that the Bhattacharyya coefficient, defined as $B(p, q) = -\log F(p, q)$, satisfies the lower bound $B(p, q) \geq \frac{1}{4} \|p - q\|_1^2$ which implies

$$F(p, q)^k \leq \exp\left(-\frac{1}{4} \|p - q\|_1^2\right). \quad (73)$$

3.3 The binomial theorem

[sec:binomial_theorem](#)

The purpose of this section is to state and prove the binomial theorem, which serves as the foundation for all subsequent results in this paper. The binomial theorem expresses any polynomial irreducible representation of a matrix sum $X + Y$ as a sum over the polynomial irreducible representations of the summands X and Y .

The theorem inherits its name from the non-commutative binomial expansion of the k -th tensor power of the sum $X + Y$,

$$(X + Y)^{\otimes k} = \sum_{\ell=0}^k \binom{k}{\ell} \left[\frac{1}{k!} \sum_{\sigma \in S_k} \sigma(X^{\otimes \ell} \otimes Y^{\otimes (k-\ell)}) \sigma^{-1} \right] \quad (74)$$

where $\sigma \in S_k$ acts on $(\mathbb{C}^n)^{\otimes k}$ by permutation of the k tensor factors. For an example, consider the case where $k = 3$:

$$\begin{aligned} (X + Y)^{\otimes 3} = & \underbrace{X \otimes X \otimes X}_{\ell=3} + \underbrace{X \otimes X \otimes Y + X \otimes Y \otimes X + Y \otimes X \otimes X}_{\ell=2} + \\ & + \underbrace{X \otimes Y \otimes Y + Y \otimes X \otimes Y + Y \otimes Y \otimes X}_{\ell=1} + \underbrace{Y \otimes Y \otimes Y}_{\ell=0}. \end{aligned} \quad (75)$$

As we have seen previously, the k -th tensor power of a matrix can also be decomposed into a sum over polynomial irreducible representations labelled by Young diagrams γ with sizes $|\gamma| = k$. Combining this formula with the non-commutative binomial expansion yields

$$(X + Y)^{\otimes k} = \sum_{\gamma \vdash k} \iota_\gamma(\pi_\gamma(X + Y) \otimes \text{id}_{W_\gamma}) \iota_\gamma^\dagger \quad (76)$$

where

$$\pi_\gamma(X + Y) = \frac{1}{\dim(W_\gamma)} \sum_{\ell=0}^k \binom{k}{\ell} \text{Tr}_{W_\gamma}[\iota_\gamma^\dagger(X^{\otimes \ell} \otimes Y^{\otimes(k-\ell)}) \iota_\gamma], \quad (77)$$

where we have eliminated the symmetrization operation over $\sigma \in S_k$ because the partial trace over W_γ is S_k -invariant. At this stage, for each summand indexed by $0 \leq \ell \leq k$, we can decompose $X^{\otimes \ell}$ into representations of X labelled by α with sizes $|\alpha| = \ell$ and $Y^{\otimes(k-\ell)}$ into representations of Y labelled by β with sizes $|\beta| = k - \ell$.

$$X^{\otimes \ell} = \sum_{\alpha \vdash \ell} \iota_\alpha(\pi_\alpha(X) \otimes \text{id}_{W_\alpha}) \iota_\alpha^\dagger, \quad Y^{\otimes(k-\ell)} = \sum_{\beta \vdash k-\ell} \iota_\beta(\pi_\beta(Y) \otimes \text{id}_{W_\beta}) \iota_\beta^\dagger. \quad (78)$$

Upon doing so, one encounters the composite operator $\iota_\gamma^\dagger(\iota_\alpha \otimes \iota_\beta)$ which maps from $(V_\alpha^n \otimes W_\alpha) \otimes (V_\beta^n \otimes W_\beta)$ to $(V_\gamma^n \otimes W_\gamma)$ in an $S_\ell \times S_{k-\ell}$ -equivariant and $U(n)$ -equivariant manner. By Schur's lemma and Schur-Weyl duality, it can be written as

$$\iota_\gamma^\dagger(\iota_\alpha \otimes \iota_\beta) = (\text{id}_{V_\gamma^n} \otimes \omega_{\alpha\beta}^\gamma)(\iota_{\alpha\beta}^\gamma \otimes \text{id}_{W_\alpha \otimes W_\beta})^\dagger \quad (79)$$

where $\iota_{\alpha\beta}^\gamma : C_{\alpha\beta}^\gamma \otimes V_\gamma^n \rightarrow V_\alpha^n \otimes V_\beta^n$ is the $U(n)$ -equivariant isometry for the γ -isotypic subspace of $V_\alpha^n \otimes V_\beta^n$ and $\omega_{\alpha\beta}^\gamma : C_{\alpha\beta}^\gamma \otimes W_\alpha \otimes W_\beta \rightarrow W_\gamma$ the $S_\ell \times S_{k-\ell}$ -equivariant isometry for the (α, β) -isotypic subspace of W_γ .⁵ Combining everything together, and rearranging we obtain the formula

$$\frac{\dim W_\gamma \pi_\gamma(X + Y)}{|\gamma|!} = \sum_{\alpha, \beta} \text{Tr}_{C_{\alpha\beta}^\gamma} \left[(\iota_{\alpha\beta}^\gamma)^\dagger \frac{\dim W_\alpha \pi_\alpha(X)}{|\alpha|!} \otimes \frac{\dim W_\beta \pi_\beta(Y)}{|\beta|!} (\iota_{\alpha\beta}^\gamma) \right], \quad (80)$$

where we leave the summation over all pairs (α, β) such that $|\alpha| = \ell$, $|\beta| = k - \ell$ for $|\gamma| = k$ implicit simply because the multiplicity space $C_{\alpha\beta}^\gamma$ is empty if $|\alpha| + |\beta| \neq |\gamma|$ and thus the corresponding summand is taken to be zero.

In the remaining of the paper, we will use the above formula, Eq. (80), to prove numerous results concerning the eigenvalues of Hermitian matrices and their sums. In order to simplify the presentation of our results considerably, we introduce the following definitions.

Definition 31 (Hook-rescaled representations). Given an $n \times n$ matrix X and a Young diagram λ , define an operator $\Phi_\lambda(X)$ acting on the representation space V_λ^n as:

$$\Phi_\lambda(X) := \frac{\pi_\lambda(X)}{H_\lambda}, \quad (81)$$

where $H_\lambda = \prod_{b \in \lambda} h_\lambda(b)$ is the product of the Hook-lengths of the boxes in the Young diagram λ . Furthermore, the trace of $\Phi_\lambda(X)$ be

$$\varphi_\lambda(X) = \text{Tr}[\Phi_\lambda(X)] = \frac{s_\lambda(X)}{H_\lambda}. \quad (82)$$

Remark 32. The operator $\Phi_\lambda(X)$ and trace $\varphi_\lambda(X)$ are simply rescalings of the representation matrix $\pi_\lambda(X)$ of X and the character $\chi_\lambda(X)$ of X respectively. This normalization scheme is such that summing the rescaled character over all Young diagrams λ with fixed size k or over all sizes yields respectively

$$\sum_{\lambda \vdash k} \varphi_\lambda(X) = \frac{\text{Tr}(X)^k}{k!}, \quad \sum_{\lambda} \varphi_\lambda(X) = \exp(\text{Tr}(X)). \quad (83)$$

⁵Schur-Weyl duality provides us with a isometric and bijective identification between the multiplicity spaces for V_γ^n inside $V_\alpha^n \otimes V_\beta^n$ with respect to $U(n)$ and $W_\alpha \otimes W_\beta$ inside W_γ with respect to $S_\ell \times S_{k-\ell}$, hence why we use, by abuse of notation, the symbol $C_{\alpha\beta}^\gamma$ in both cases.

Before stating the main result of this section, we first generalize the quantities $\Phi_\lambda(X)$ and $\varphi_\lambda(X)$ to consider arguments involving more than one matrix.

Definition 33 (Mixed hook-rescaled representations). Given a pair (X, Y) of $n \times n$ matrices and a triple of Young diagrams (α, β, γ) , define an operator $\Phi_\gamma^{\alpha\beta}(X, Y)$ acting on the representation space V_γ^n as

$$\Phi_\gamma^{\alpha\beta}(X, Y) := \text{Tr}_{C_{\alpha,\beta}^\gamma} [\iota_{\alpha,\beta}^{\gamma;\dagger}(\Phi_\alpha(X) \otimes \Phi_\beta(Y)) \iota_{\alpha,\beta}^\gamma], \quad (84)$$

where $\iota_{\alpha,\beta}^\gamma : C_{\alpha,\beta}^\gamma \otimes V_\gamma^n \rightarrow V_\alpha^n \otimes V_\beta^n$ is the $U(n)$ -equivariant isometry for the γ -isotypic subspace of $V_\alpha^n \otimes V_\beta^n$. Note the sum over Young diagrams α , and β is implicitly restricted to those satisfying $|\alpha| + |\beta| = |\gamma|$, since otherwise the multiplicity space $C_{\alpha,\beta}^\gamma$ is zero-dimensional and thus the summand is taken to be zero. Analogously, let the trace of $\Phi_\gamma^{\alpha\beta}(X, Y)$ be denoted by

$$\varphi_\gamma^{\alpha\beta}(X, Y) := \text{Tr}[\Phi_\gamma^{\alpha\beta}(X, Y)] = \text{Tr}[P_{\alpha,\beta}^\gamma(\Phi_\alpha(X) \otimes \Phi_\beta(Y))], \quad (85)$$

where $P_{\alpha,\beta}^\gamma = \iota_{\alpha,\beta}^{\gamma;\dagger} \iota_{\alpha,\beta}^\gamma$ is the projection operator onto the γ -isotypic subspace of $V_\alpha^n \otimes V_\beta^n$.

Remark 34. While $\Phi_\lambda(X)$ is a homogeneous polynomial in the matrix coefficients of X of degree $|\lambda|$ in the sense that $\Phi_\lambda(sX) = s^{|\lambda|} \Phi_\lambda(X)$ holds for any $s \in \mathbb{R}$, $\Phi_\gamma^{\alpha\beta}(X, Y)$ is a bihomogeneous polynomial in X and Y of degrees $|\alpha|$ and $|\beta|$ respectively in the sense that for any $s, t \in \mathbb{R}$,

$$\Phi_\gamma^{\alpha\beta}(sX, tY) = s^{|\alpha|} t^{|\beta|} \Phi_\gamma^{\alpha\beta}(X, Y). \quad (86)$$

Remark 35. The quantity $\Phi_\gamma^{\alpha\beta}(X, Y)$ generalizes $\Phi_\lambda(X)$ in the sense that one can take the triple $(\alpha, \beta, \gamma) = (\lambda, (), \lambda)$ (where $()$ denotes the Young diagram with zero boxes), in which case $V_{()}^n$ supports the trivial one-dimensional representation with $\Phi_{()}(Y) = 1$ and because $\dim(C_{\lambda,()}^\lambda) = 1$,

$$\forall Y : \quad \Phi_\lambda^{\lambda,()}(X, Y) = \Phi_\lambda(X). \quad (87)$$

With these definitions in place, we are in a position to state the main result of this section. thm:binomial

Theorem 36 (Binomial theorem). *For all $n \times n$ matrices X, Y , and all Young diagrams γ ,*

$$\Phi_\gamma(X + Y) = \sum_{\alpha,\beta} \Phi_\gamma^{\alpha\beta}(X, Y), \quad (88)$$

where the sum is implicitly taken over all Young diagrams such that $|\alpha| + |\beta| = |\gamma|$.

Proof. See preceding discussion. □

cor:bisummation

Corollary 37. *Let X, Y be $n \times n$ matrices. Then the following identities hold:*

$$\forall \alpha, \beta : \quad \sum_{\gamma} \varphi_\gamma^{\alpha\beta}(X, Y) = \varphi_\alpha(X) \varphi_\beta(Y), \quad (89)$$

$$\forall \gamma : \quad \sum_{\alpha,\beta} \varphi_\gamma^{\alpha\beta}(X, Y) = \varphi_\gamma(X + Y). \quad (90)$$

Proof. The first identity involving a summation over α and β is simply the result of taking the trace of both sides of Theorem 36. For the second identity, involving a summation over γ , follows from the resolution of the identity $\sum_{\gamma \vdash |\alpha| + |\beta|} P_{\alpha,\beta}^\gamma = \text{id}_{V_\alpha^n \otimes V_\beta^n}$:

$$\varphi_\alpha(X) \varphi_\beta(Y) = \text{Tr}[\Phi_\alpha(X) \otimes \Phi_\beta(Y)] = \sum_{\gamma} \text{Tr}[P_{\alpha,\beta}^\gamma(\Phi_\alpha(X) \otimes \Phi_\beta(Y))] = \sum_{\gamma} \varphi_\gamma^{\alpha\beta}(X, Y), \quad (91)$$

where the last equality is just the definition of $\varphi_\gamma^{\alpha\beta}(X, Y)$. Once again, the condition that $|\gamma| = |\alpha| + |\beta|$ is implicitly present. □

Remark 38. One might consider what happens to the operator $\Phi_\gamma^{\alpha\beta}(X, Y)$ when one sums over γ instead of (α, β) . However, an analogous identity concerning a sum over γ of the operator $\Phi_\gamma^{\alpha\beta}(X, Y)$ does not immediately follow because the tensor product $\Phi_\alpha(X) \otimes \Phi_\beta(Y)$ is not necessarily block diagonalizable into γ -isotypic subspaces in any obvious way. One exception occurs when $X = Y$, in which case,

$$\pi_\alpha(X) \otimes \pi_\beta(X) = \sum_\gamma \iota_{\alpha\beta}^\gamma (\pi_\gamma(X) \otimes \text{id}_{C_{\alpha\beta}^\gamma}) \iota_{\alpha\beta}^{\gamma;\dagger}, \quad (92)$$

which is the operator version of the character product formula $\chi_\alpha(X)\chi_\beta(X) = \sum_\gamma c_{\alpha\beta}^\gamma \chi_\gamma(X)$.

3.4 Inequalities for Horn's original problem

sec:new_ineq_horn_orig

In Section 3.3 we introduced to the quantity $\varphi_\gamma^{\alpha\beta}(X, Y)$ and saw in Corollary 37 that it satisfied two summation identities. Under the assumption that the matrices X , and Y are positive semidefinite, we can conclude that $\varphi_\gamma^{\alpha\beta}(X, Y)$ is non-negative. This observation implies the following upper bounds on $\varphi_\gamma^{\alpha\beta}(X, Y)$.

lem:XY_marginal_upper_bounds

Lemma 39. *Let $X, Y \geq 0$ be positive semidefinite $n \times n$ matrices. For all Young diagrams (α, β, γ) , we have the inequality*

$$\varphi_\gamma^{\alpha\beta}(X, Y) \leq \min\{\varphi_\alpha(X)\varphi_\beta(Y), \varphi_\gamma(X + Y)\}. \quad (93)$$

Proof. The claim follows directly from Corollary 37 provided $\varphi_\gamma^{\alpha\beta}(X, Y) \geq 0$. However, this follows because $\varphi_\gamma^{\alpha\beta}(X, Y) = \text{Tr}[P_{\alpha\beta}^\gamma(\Phi_\alpha(X) \otimes \Phi_\beta(Y))]$ and $P_{\alpha\beta}^\gamma \geq 0$ (it is a projection operator) and $\Phi_\alpha(X), \Phi_\beta(Y) \geq 0$ because $X, Y \geq 0$. $\square$

cor:orig_horn_ineq

Corollary 40. *Let $X, Y \geq 0$ be positive semidefinite $n \times n$ matrices. Then the following inequalities hold:*

$$\forall \alpha, \beta : \quad \varphi_\alpha(X)\varphi_\beta(Y) \leq \sum_{\substack{\alpha, \beta \\ c_{\alpha\beta}^\gamma > 0}} \varphi_\gamma(X + Y), \quad (94)$$

$$\forall \gamma : \quad \varphi_\gamma(X + Y) \leq \sum_{\substack{\alpha, \beta \\ c_{\alpha\beta}^\gamma > 0}} \varphi_\alpha(X)\varphi_\beta(Y). \quad (95)$$

Proof. The claim follows by applying upper bounds from Lemma 39 to the summations in Corollary 37 being careful to note that if $c_{\alpha\beta}^\gamma = 0$, then $\varphi_\gamma^{\alpha\beta}(X, Y) = 0$ and one only needs to consider the cases where $c_{\alpha\beta}^\gamma \neq 0$. $\square$

3.5 The $6j$ symbol for $U(n)$

sec:sixj_tet

The purpose of this section is to introduce a representation theoretic object, namely the $6j$ symbol, and then in the subsequent sections show how it places constraints on whether a given sextuple of eigenvalues (a, b, c, d, e, f) is tetrahedral in a manner which generalizes how the Littlewood-Richardson coefficient encodes whether a given triple (a, b, c) is trigonal. For the case where $n = 2$, this correspondence is well-known [Rob99; PR68; Bac10], while for the case where $n \geq 3$, this correspondence was already established indirectly by Christandl, Şahinoğlu, and Walter [CŞW18]. Here we establish this correspondence directly.

As we discussed in the introduction, the $6j$ symbol emerges naturally when considering successive decompositions of the triple tensor product $V_\alpha^n \otimes V_\beta^n \otimes V_\delta^n$.

defn:sixj

Definition 41. Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be a sextuple of Young diagrams with sizes satisfying

$$|\alpha| + |\beta| = |\gamma|, \quad |\beta| + |\delta| = |\phi|, \quad |\gamma| + |\delta| = |\epsilon|, \quad |\alpha| + |\phi| = |\epsilon|. \quad (96)$$

The $U(n)$ - $6j$ operator is the operator⁶

$$\left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} : C_{\alpha, \beta}^\gamma \otimes C_{\delta, \gamma}^\epsilon \rightarrow C_{\alpha, \phi}^\epsilon \otimes C_{\beta, \delta}^\phi, \quad (97)$$

defined by

$$\left\{ \begin{array}{ccc} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{array} \right\} := \frac{1}{\dim(V_\epsilon)} \text{Tr}_{V_\epsilon} [\iota_{\alpha\phi}^{\epsilon;\dagger} (\text{id}_{V_\alpha^n} \otimes \iota_{\beta\delta}^{\phi;\dagger}) (\iota_{\alpha\beta}^\gamma \otimes \text{id}_{V_\delta^n}) \iota_{\gamma\delta}^\epsilon], \quad (\text{eq:sixj}) \quad (98)$$

and depicted diagrammatically by

$$\left\{ \begin{array}{ccc} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{array} \right\} := \frac{1}{(V_\epsilon)} \text{Tr}_{V_\epsilon} [\iota_{\alpha\phi}^{\epsilon;\dagger} (\text{id}_{V_\alpha^n} \otimes \iota_{\beta\delta}^{\phi;\dagger}) (\iota_{\alpha\beta}^\gamma \otimes \text{id}_{V_\delta^n}) \iota_{\gamma\delta}^\epsilon] = \frac{1}{(V_\epsilon)} \text{Tr}_{V_\epsilon} [V_\alpha V_\beta V_\gamma V_\delta V_\epsilon] \quad (99)$$

To illustrate the situation more concretely, consider the following three projection operators on the triple tensor product $V_\alpha^n \otimes V_\beta^n \otimes V_\delta^n$:

$$P_{\alpha\beta}^\gamma \otimes \text{id}_{V_\delta^n}, \quad P_{\alpha\beta\delta}^\epsilon, \quad \text{id}_{V_\alpha^n} \otimes P_{\beta\delta}^\phi. \quad (100)$$

In general, $P_{\alpha\beta}^\gamma$ and $P_{\beta\delta}^\phi$ do not necessarily commute with each other in the sense that for some values of $\alpha, \beta, \delta, \gamma$, and ϕ , one observes

$$[P_{\alpha\beta}^\gamma \otimes \text{id}_{V_\delta^n}, \text{id}_{V_\alpha^n} \otimes P_{\beta\delta}^\phi] \neq 0. \quad (101)$$

Nevertheless, the projection operators $P_{\alpha\beta}^\gamma$ and $P_{\beta\delta}^\phi$ each respectively commute with $P_{\alpha\beta\delta}^\epsilon$ in the sense that for all $\alpha, \beta, \gamma, \delta, \epsilon$, and ϕ ,

$$[P_{\alpha\beta\delta}^\epsilon, P_{\alpha\beta}^\gamma \otimes \text{id}_{V_\delta^n}] = 0, \quad [P_{\alpha\beta\delta}^\epsilon, \text{id}_{V_\alpha^n} \otimes P_{\beta\delta}^\phi] = 0. \quad (102)$$

Let $\Pi_{\alpha\beta\delta}^{\epsilon;\gamma}$ and $\Pi_{\alpha\beta\delta}^{\epsilon;\phi}$ be projection operators given by

$$\Pi_{\alpha\beta\delta}^{\epsilon;\gamma} := P_{\alpha\beta\delta}^\epsilon (\text{id}_{V_\alpha^n} \otimes P_{\beta\delta}^\phi), \quad \Pi_{\alpha\beta\delta}^{\epsilon;\phi} := P_{\alpha\beta\delta}^\epsilon (\text{id}_{V_\alpha^n} \otimes P_{\beta\delta}^\gamma). \quad (\text{eq:double_projs}) \quad (103)$$

prop:6j_norm_double_proj

Proposition 42. For any sextuple of Young diagrams $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$,

$$\left\| \left\{ \begin{array}{ccc} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{array} \right\} \right\|_\infty = \left\| \Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \Pi_{\alpha\beta\delta}^{\epsilon;\phi} \right\|_\infty, \quad \left\| \left\{ \begin{array}{ccc} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{array} \right\} \right\|_2^2 = \frac{1}{\dim V_\epsilon^n} \text{Tr} [\Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \Pi_{\alpha\beta\delta}^{\epsilon;\phi}]. \quad (104)$$

3.6 Inequalities for the tetrahedral Horn problem

sec:new_ineq_horn_tet

Definition 43. Let X, Y, Z be $n \times n$ matrices. For any sextuple $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ of Young diagrams, let

$$\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) := \text{Tr} [(P_{\alpha\beta}^\gamma \otimes \text{id}_{V_\delta^n}) P_{\alpha\beta\delta}^\epsilon (\text{id}_{V_\alpha^n} \otimes P_{\beta\delta}^\phi) \Phi_\alpha(X) \otimes \Phi_\beta(Y) \otimes \Phi_\delta(Z)]. \quad (105)$$

⁶It is important to note that there does not seem to exist a universal convention concerning how 6j symbols are normalized; Coquereaux discusses this problem in [Coq23].

$$\begin{array}{ccccccc}
& & \xrightarrow{\Sigma_\gamma} & \varphi_{\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) & \xrightarrow{\Sigma_\epsilon} & \varphi_\alpha(X)\varphi_\phi^{\beta\delta}(Y, Z) & \xrightarrow{\Sigma_{\beta\delta}} \varphi_\alpha(X)\varphi_\phi(Y+Z) \\
\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) & \xrightarrow{\Sigma_{\alpha\beta\delta\gamma\phi}} & \varphi_\epsilon(X+Y+Z) & \xleftarrow{\Sigma_{\alpha\beta\delta}} & \varphi_\epsilon^{\alpha\beta\delta}(X, Y, Z) & \xrightarrow{\Sigma_\epsilon} & \varphi_\alpha(X)\varphi_\beta(Y)\varphi_\delta(Z) \\
& & \xrightarrow{\Sigma_\phi} & \varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z) & \xrightarrow{\Sigma_\epsilon} & \varphi_{\gamma-}^{\alpha\beta}(X, Y)\varphi_\delta(Z) & \xrightarrow{\Sigma_{\alpha\beta}} \varphi_\gamma(X+Y)\varphi_\delta(Z)
\end{array}$$

$\downarrow \Sigma_\phi$ $\downarrow \Sigma_\phi$
 $\uparrow \Sigma_\gamma$ $\uparrow \Sigma_\gamma$

Figure 3: Various summation identities satisfied by $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$.

`fig:sum_ident`

Remark that in our definition of $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$, we permit the sizes of the involved Young diagrams to be arbitrary. However, there are numerous conditions on the sizes of the Young diagrams $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ which are necessary (but not always sufficient) for $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ to be non-zero; specifically,

$$|\alpha| + |\beta| = |\gamma|, \quad |\beta| + |\delta| = |\phi|, \quad |\alpha| + |\phi| = |\epsilon|, \quad |\gamma| + |\delta| = |\epsilon|. \quad (106)$$

`{eq:yd_sizes_tet}`

If we additionally let $\Omega_{\alpha\beta\delta}^{X,Y,Z} := \Phi_\alpha(X) \otimes \Phi_\beta(Y) \otimes \Phi_\delta(Z)$, then we can compactly express $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ as

$$\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) = \text{Tr}[\Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \Pi_{\alpha\beta\delta}^{\epsilon;\phi} \Omega_{\alpha\beta\delta}^{X,Y,Z}]. \quad (107)$$

`{eq:varphi_compact_tet}`

Unlike the quantity $\varphi_{\gamma\beta}^{\alpha\beta}(X, Y) = \text{Tr}[P_{\alpha\beta}^\gamma \Phi_\alpha(X) \otimes \Phi_\beta(Y)]$ involving two matrices X, Y (see Section 3.3), the quantity defined here, $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$, is not necessarily nonnegative or even real valued even under the assumption that $X, Y, Z \geq 0$ are positive semidefinite.

Nevertheless, summing over either γ or ϕ restores non-negativity because it eliminates one of the potentially non-commuting projection operators $\Pi_{\alpha\beta\delta}^{\epsilon;\gamma}$ or $\Pi_{\alpha\beta\delta}^{\epsilon;\phi}$. We denote these parital sums of $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ over γ and ϕ respectively as

$$\varphi_{\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) := \sum_{\gamma} \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) = \text{Tr}[\Pi_{\alpha\beta\delta}^{\epsilon;\phi} \Omega_{\alpha\beta\delta}^{X,Y,Z}] = \text{Tr}[P_{\gamma\delta}^\epsilon(\Phi_\gamma^{\alpha\beta}(X, Y) \otimes \Phi_\delta(Z))], \quad (108)$$

$$\varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z) := \sum_{\phi} \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) = \text{Tr}[\Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \Omega_{\alpha\beta\delta}^{X,Y,Z}] = \text{Tr}[P_{\alpha\phi}^\epsilon(\Phi_\alpha(X) \otimes \Phi_\phi^{\beta\delta}(Y, Z))], \quad (109)$$

where the sum over γ is implicitly such that $|\gamma| = |\alpha| + |\beta|$ and the sum over ϕ is implicitly such that $|\phi| = |\beta| + |\delta|$. Although $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ is not necessarily real or positive, the above expression makes it evident that

$$\varphi_{\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \geq 0, \quad \varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z) \geq 0. \quad (110)$$

The various summation identities satisfied by Fig. 3 are summarized in Fig. 3.

The next series of results concerns upper bounds on the absolute value of $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$. These upper bounds follow from the following general lemma applied to the compact expression for $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ appearing in Eq. (107).

`lem:cs_ineq_double_proj`

Lemma 44. *Let $\Omega \geq 0$ be a positive semidefinite operator and let Π_1, Π_2 be Hermitian projection operators acting on a common vector space. Then*

$$|\text{Tr}(\Pi_1 \Pi_2 \Omega)| \leq \|\Pi_1 \Pi_2\|_\infty \text{Tr}(\Pi_1 \Omega)^{\frac{1}{2}} \text{Tr}(\Pi_2 \Omega)^{\frac{1}{2}}. \quad (111)$$

Proof. As $\Omega \geq 0$ is positive semidefinite, let $\Omega^{\frac{1}{2}} \geq 0$ denote its square root. Then let $L = \Pi_1 \Pi_2 \Omega^{\frac{1}{2}}$ and $R = \Pi_1 \Omega^{\frac{1}{2}}$ so that

$$\langle L, R \rangle = \text{Tr}(L^\dagger R) = \text{Tr}(\Pi_1 \Pi_2 \Omega \Pi_1) = \text{Tr}(\Pi_1 \Pi_2 \Omega), \quad (112)$$

where the last equality uses the assumption that $\Pi_1^2 = \Pi_1$. An application of the Cauchy-Schwarz inequality for the Hilbert-Schmidt inner product $\langle L, R \rangle$ then implies

$$|\text{Tr}(\Pi_1 \Pi_2 \Omega)|^2 = |\langle L, R \rangle|^2 \leq \text{Tr}[L^\dagger L] \text{Tr}[R^\dagger R] = \text{Tr}[\Pi_1 \Pi_2 \Omega \Pi_2 \Pi_1] \text{Tr}[\Pi_1 \Omega \Pi_1] = \text{Tr}[\Pi_1 \Pi_2 \Omega \Pi_2] \text{Tr}[\Pi_1 \Omega]. \quad (113)$$

Now let $\{v_j\}_j$ be an orthonormal eigenbasis for $\Pi_2\Omega\Pi_2$ with eigenvalue $\omega_j \geq 0$. Then, because it was assumed that $\Pi_2^2 = \Pi_2$, we have

$$\text{Tr}[\Pi_1\Pi_2\Omega\Pi_2] = \text{Tr}[(\Pi_2\Pi_1^2\Pi_2)(\Pi_2\Omega\Pi_2)] = \sum_j \omega_j \langle \Pi_1\Pi_2 v_j, \Pi_1\Pi_2 v_j \rangle \leq \sum_j \omega_j \|\Pi_1\Pi_2\|_\infty^2 = \text{Tr}[\Pi_2\Omega] \|\Pi_1\Pi_2\|_\infty^2. \quad (114)$$

The claim then follows from combining these inequalities and taking the square root of both sides. $\square$ thm:varphi_tet_main_ineq

Theorem 45. *Let $X, Y, Z \geq 0$ be positive semidefinite $n \times n$ matrices. Then for all sextuples of Young diagrams $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$,*

$$\left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \right| \leq \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_\infty \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)^{\frac{1}{2}} \varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z)^{\frac{1}{2}}. \quad (115)$$

Proof. Under the assumption that $X, Y, Z \geq 0$, we have that $\Omega_{\alpha\beta\delta}^{X,Y,Z} = \Phi_\alpha(X) \otimes \Phi_\beta(Y) \otimes \Phi_\delta(Z) \geq 0$ is positive semidefinite also. Now applying Lemma 44 to the compact expression for $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ in Eq. (107) yields

$$\left| \text{Tr}[\Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \Pi_{\alpha\beta\delta}^{\epsilon;\phi} \Omega_{\alpha\beta\delta}^{X,Y,Z}] \right| \leq \left\| \Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \Pi_{\alpha\beta\delta}^{\epsilon;\phi} \right\|_\infty \text{Tr}[\Pi_{\alpha\beta\delta}^{\epsilon;\gamma} \Omega_{\alpha\beta\delta}^{X,Y,Z}]^{\frac{1}{2}} \text{Tr}[\Pi_{\alpha\beta\delta}^{\epsilon;\phi} \Omega_{\alpha\beta\delta}^{X,Y,Z}]^{\frac{1}{2}} \quad (116)$$

which is equivalent to the claim due to Proposition 42. $\square$

lem:tet_partial_sum_ineq

Lemma 46. *Let $X, Y, Z \geq 0$ be positive semidefinite $n \times n$ matrices. Then we have bounds on the partial sums,*

$$\begin{aligned} \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) &\leq \varphi_\alpha(X) \varphi_\phi(Y + Z), \\ \varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z) &\leq \varphi_\gamma(X + Y) \varphi_\delta(Z), \end{aligned} \quad (117)$$

as well as bounds which are uniform in both γ and ϵ :

$$\max\{\varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z), \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)\} \leq \varphi_\epsilon^{\alpha\beta\delta}(X, Y, Z) \leq \min\{\varphi_\epsilon(X + Y + Z), \varphi_\alpha(X) \varphi_\beta(Y) \varphi_\delta(Z)\}. \quad (118)$$

Proof. The bounds are consequences from summation identities and the fact that the partial sums $\varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ and $\varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z)$ are non-negative real-valued. The first pair of bounds follow from the summation identities

$$\begin{aligned} \varphi_\alpha(X) \varphi_\phi(Y + Z) &= \sum_{\epsilon\beta\delta} \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z), \\ \varphi_\gamma(X + Y) \varphi_\delta(Z) &= \sum_{\epsilon\alpha\beta} \varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z). \end{aligned} \quad (119)$$

The last inequality comes from

$$\varphi_\epsilon^{\alpha\beta\delta}(X, Y, Z) = \sum_\phi \varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) = \sum_\gamma \varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z) \quad (120)$$

and the fact that

$$\begin{aligned} \varphi_\alpha(X) \varphi_\beta(Y) \varphi_\delta(Z) &= \sum_\epsilon \varphi_\epsilon^{\alpha\beta\delta}(X, Y, Z), \\ \varphi_\epsilon(X + Y + Z) &= \sum_{\alpha\beta\delta} \varphi_\epsilon^{\alpha\beta\delta}(X, Y, Z). \end{aligned} \quad (121)$$

cor:tet_eigenvalue_cube $\square$

Corollary 47. *Let $X, Y, Z \geq 0$ be positive semidefinite $n \times n$ matrices. Then for all triples of Young diagrams (α, β, δ) ,*

$$\varphi_\alpha(X)^{\frac{2}{3}} \varphi_\beta(Y) \varphi_\delta(Z)^{\frac{2}{3}} \leq \sum_{\gamma, \epsilon, \phi} \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_\infty \varphi_\phi(Y + Z)^{\frac{1}{3}} \varphi_\gamma(X + Y)^{\frac{1}{3}} \varphi_\epsilon(X + Y + Z)^{\frac{1}{3}}. \quad (122)$$

Proof. The claim follows by summing the inequality in Theorem 45 over triples (γ, ϵ, ϕ) and relaxing both the sides of the resulting inequality. For the lower bound, we have $\varphi_\alpha(X)\varphi_\beta(Y)\varphi_\delta(Z) = \sum_{\gamma\epsilon\phi} \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)$ which implies

$$\varphi_\alpha(X)\varphi_\beta(Y)\varphi_\delta(Z) \leq \sum_{\gamma\epsilon\phi} \left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \right|. \quad (123)$$

Then for the upper bound, Lemma 46 can be used to produce

$$\varphi_{-\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)\varphi_{\gamma\epsilon-}^{\alpha\beta\delta}(X, Y, Z) \leq [\varphi_\alpha(X)\varphi_\phi(Y+Z)\varphi_\gamma(X+Y)\varphi_\delta(Z)]^p \varphi_\epsilon(X+Y+Z)^{2(1-p)}, \quad (124)$$

which holds for any $p \in [0, 1]$. The claim corresponds to the choice $p = \frac{2}{3}$ which makes the exponents match. Then the claimed result is obtained from dividing out both sides by a common factor of $(\varphi_\alpha(X)\varphi_\epsilon(Y))^{\frac{1}{3}}$ (note that if $\varphi_\alpha(X)\varphi_\epsilon(Y) = 0$, the inequality would hold trivially). $\square$

3.7 Results for random matrices

sec:random_matrices

The purpose of this section is record some results concerning sums of random Hermitian matrices with fixed eigenvalues sampled uniformly from their respective orbits. The first two propositions follow from standard applications of Schurs lemma, and are used to prove the others.

Proposition 48. *Let $x = (x_1 \geq \dots \geq x_n)$ be a set of eigenvalues and let $X \sim \mathcal{O}_x$ denote a random Hermitian matrix of the form $X = U[x]U^{-1}$ where $U \in \mathbf{U}(n)$ is sampled uniformly with respect to the normalized Haar measure on $\mathbf{U}(n)$. Then for any Young diagram λ ,*

$$\mathbb{E}_{X \sim \mathcal{O}_x} \pi_\lambda(X) = s_\lambda(x) \frac{\text{id}_{V_\lambda^n}}{\dim V_\lambda^n}. \quad (125)$$

Proposition 49. *Given a positive real number $\tau > 0$, let $X \sim \text{HS}_\tau$ denote a random Hermitian matrix with coefficients $X_{ij} = \frac{\tau}{n} \sum_{k,s,t=1}^n U_{ik;ss} U_{tt;jk}^{-1}$ where $U \in \mathbf{U}(n^2)$ is a $n^2 \times n^2$ unitary matrix sampled uniformly with respect to the normalized Haar measure on $\mathbf{U}(n^2)$. Then for any Young diagram λ ,*

$$\mathbb{E}_{X \sim \text{HS}_\tau} \pi_\lambda(X) = \tau^{|\lambda|} q_{n,\lambda} \frac{\text{id}_{V_\lambda^n}}{\dim V_\lambda^n}. \quad (126)$$

cor:random_sum

Corollary 50. *Let (x, y) be a pair of eigenvalues. Then*

$$\mathbb{E}_{X \sim \mathcal{O}_x} \mathbb{E}_{Y \sim \mathcal{O}_y} \frac{\varphi_\gamma^{\alpha\beta}(X, Y)}{\dim V_\gamma^n} = c_{\alpha\beta}^\gamma \frac{\varphi_\alpha(x)}{\dim V_\alpha^n} \frac{\varphi_\beta(y)}{\dim V_\beta^n}. \quad (127)$$

$$\mathbb{E}_{X \sim \text{HS}_{\text{Tr}[x]}} \mathbb{E}_{Y \sim \text{HS}_{\text{Tr}[y]}} \varphi_\gamma^{\alpha\beta}(X, Y) = \frac{\text{Tr}[x]^{|\alpha|}}{|\alpha|!} \frac{\text{Tr}[y]^{|\beta|}}{|\beta|!} q_{n,\alpha} q_{n,\beta} \frac{c_{\alpha\beta}^\gamma \dim V_\gamma^n}{\dim(V_\alpha^n \otimes V_\beta^n)}. \quad (128)$$

cor:random_matrices_tet

Corollary 51. *Let (x, y, z) be a triple of eigenvalues. Then*

$$\mathbb{E}_{X \sim \mathcal{O}_x} \mathbb{E}_{Y \sim \mathcal{O}_y} \mathbb{E}_{Z \sim \mathcal{O}_z} \frac{\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z)}{\dim V_\epsilon^n} = \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_2^2 \frac{\varphi_\alpha(x)}{\dim V_\alpha^n} \frac{\varphi_\beta(y)}{\dim V_\beta^n} \frac{\varphi_\delta(z)}{\dim V_\delta^n}. \quad (129)$$

$$\mathbb{E}_{X \sim \text{HS}_{\text{Tr}[x]}} \mathbb{E}_{Y \sim \text{HS}_{\text{Tr}[y]}} \mathbb{E}_{Z \sim \text{HS}_{\text{Tr}[z]}} \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) = \frac{\text{Tr}[x]^{|\alpha|}}{|\alpha|!} \frac{\text{Tr}[y]^{|\beta|}}{|\beta|!} \frac{\text{Tr}[z]^{|\delta|}}{|\delta|!} q_{n,\alpha} q_{n,\beta} q_{n,\delta} \frac{\dim V_\epsilon^n \left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_2^2}{\dim(V_\alpha^n \otimes V_\beta^n \otimes V_\delta^n)}. \quad (130)$$

3.8 Proof of Theorem 3

To begin we use the random matrix identities from Section 3.7 to prove that $\varphi_\gamma(A + B)$ does not differ significantly when A, B are replaced with random matrices in their respective coadjoint orbits. This result follows from Corollary 40.

cor:random_sum_ineq

Corollary 52. *For a given γ , let $m_n(\gamma)$ be the following quantity*

$$m_n(\gamma) := \min_{\substack{\alpha, \beta \\ c_{\alpha\beta}^\gamma > 0}} \frac{c_{\alpha\beta}^\gamma \dim V_\gamma^n}{\dim V_\alpha^n \otimes V_\beta^n} \quad (131)$$

Then for all $A, B \geq 0$,

$$\varphi_\gamma(A + B) \leq \sum_{\substack{\alpha, \beta \\ c_{\alpha\beta}^\gamma > 0}} \varphi_\alpha(A) \varphi_\beta(B) \leq \frac{1}{m_n(\gamma)} \mathbb{E}_{A' \sim \mathcal{O}_a} \mathbb{E}_{B' \sim \mathcal{O}_b} \varphi_\gamma(A' + B') \quad (132)$$

Before continuing to the proof of Theorem 3, we remark that the prefactor $\frac{c_{\alpha\beta}^\gamma \dim V_\gamma^n}{\dim V_\alpha^n \otimes V_\beta^n}$ above can be viewed as a probability distribution over Young diagrams γ of size $|\gamma| = |\alpha| + |\beta|$ when conditioned on fixed α, β because it corresponds to the proportion of $V_\alpha^n \otimes V_\beta^n$ occupied by the γ -isotypic subspace:

$$\sum_{\gamma} c_{\alpha\beta}^\gamma \dim V_\gamma^n = \dim(V_\alpha^n \otimes V_\beta^n). \quad (133)$$

From this point of view, the quantity $m_n(\gamma)$ can be interpreted as the minimum likelihood for the outcome γ as one varies over all pairs of Young diagrams (α, β) of sizes fixed by γ .

Proof of Theorem 3. That Eq. (4) must be satisfied by (a, b, c) whenever there exists follows from Corollary 40. To prove the bound on $D(a, b, c)$, it actually suffices for (a, b, c) to satisfy a weaker constraint. Specifically, the requirement that Eq. (4) is satisfied for a given γ can be relaxed, by Corollary 52, to the requirement that

$$\varphi_\gamma(c) \leq \frac{1}{m_n(\gamma)} \max_{(X, Y) \in \mathcal{O}_a \times \mathcal{O}_b} \varphi_\gamma(X + Y). \quad (134)$$

Since it was assumed that $\text{Tr}[c] = \text{Tr}[a] + \text{Tr}[b] = \text{Tr}(X)$, an application of the eigenvalue separation lemma produces:

$$\varphi_\gamma(c) = \sqrt{\varphi_\gamma(c)} \sqrt{\varphi_\gamma(c)}, \quad (135)$$

$$\leq \sqrt{\frac{1}{m_n(\gamma)}} \max_{(X, Y) \in \mathcal{O}_a \times \mathcal{O}_b} \sqrt{\varphi_\gamma(c) \varphi_\gamma(X + Y)}, \quad (136)$$

$$\leq \frac{\text{Tr}[c]^k}{k!} \sqrt{\frac{(\dim V_\gamma^n)^2}{m_n(\gamma)}} \exp\left(-\frac{k}{4 \text{Tr}[c]^2} D(a, b, c)^2\right). \quad (137) \quad \text{\texttt{eq:eig_sep_lr}}$$

Furthermore, note that

$$\frac{(\dim V_\gamma^n)^2}{m_n(\gamma)} = \max_{\substack{\alpha, \beta \\ c_{\alpha\beta}^\gamma > 0}} \frac{\dim V_\alpha^n \otimes V_\beta^n \otimes V_\gamma^n}{c_{\alpha\beta}^\gamma} \leq \left((k+1)^{n(n-1)/2}\right)^3. \quad (138)$$

Since this constraint is assumed to hold for all $\gamma \vdash k$, we may sum over all $\gamma \vdash k$. After summing and dividing both sides of Eq. (137) by $\frac{\text{Tr}[c]^k}{k!}$ (and noting there are at most $(k+1)^{n-1}$ summands) yields

$$1 \leq (k+1)^{3n(n-1)/4+n-1} \exp\left(-\frac{k}{4 \text{Tr}[c]^2} D(a, b, c)^2\right). \quad (139)$$

Rearranging for $D(a, b, c)$ then yields the claimed bound:

$$\frac{1}{\text{Tr}[c]} D(a, b, c) \leq \sqrt{(3n^2 + n - 4) \frac{\log(k+1)}{k}}. \quad (140)$$

□

3.9 Proof of Theorem 1

Proof of Theorem 1. If a sextuple of eigenvalues (a, b, c, d, e, f) happens to satisfy Eq. (2) for a particular triple of Young diagrams (α, β, δ) , then multiplying both sides of Eq. (2) by $\varphi_\alpha(a)^{\frac{4}{3}} \varphi_\beta(b) \varphi_\delta(d)^{\frac{4}{3}}$ yields

$$(\varphi_\alpha(a) \varphi_\beta(b) \varphi_\delta(d))^2 \leq \sum_{\gamma, \epsilon, \phi} \left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_\infty \varphi_\alpha(a)^{\frac{4}{3}} \varphi_\beta(b) \varphi_\delta(d)^{\frac{4}{3}} \varphi_\phi(f)^{\frac{1}{3}} \varphi_\gamma(c)^{\frac{1}{3}} \varphi_\epsilon(e)^{\frac{1}{3}}. \quad (141)$$

Our objective is to sum this expression over all triples (α, β, δ) such that $|\alpha| + |\beta| + |\delta| = k$. Note that summing over $\varphi_\alpha(a) \varphi_\beta(b) \varphi_\delta(d)$ yields

$$\sum_{\substack{\alpha, \beta, \delta \\ |\alpha| + |\beta| + |\delta| = k}} \varphi_\alpha(a) \varphi_\beta(b) \varphi_\delta(d) = \frac{\text{Tr}[e]^k}{k!} \quad (142)$$

since we explicitly assumed the trace conditions held, i.e., $\text{Tr}[e] = \text{Tr}[a] + \text{Tr}[b] + \text{Tr}[d]$.

For notational convenience, let $W_k \left(\begin{smallmatrix} a & b & c \\ d & e & f \end{smallmatrix} \right)$ denote the corresponding sum of the right-hand side (appropriately normalized by $\frac{\text{Tr}[e]^k}{k!}$ so that W_k is invariant under rescaling):

$$W_k \left(\begin{smallmatrix} a & b & c \\ d & e & f \end{smallmatrix} \right) := \left(\frac{k!}{\text{Tr}[e]^k} \right)^2 \sum_{\substack{\alpha, \beta, \delta \\ \gamma, \epsilon, \phi \\ |\epsilon| = k}} \left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_\infty \varphi_\alpha(a)^{\frac{4}{3}} \varphi_\beta(b) \varphi_\delta(d)^{\frac{4}{3}} \varphi_\phi(f)^{\frac{1}{3}} \varphi_\gamma(c)^{\frac{1}{3}} \varphi_\epsilon(e)^{\frac{1}{3}}. \quad (143)$$

The bound in Eq. (141) then implies a moment inequality bound:

$$W_k \left(\begin{smallmatrix} a & b & c \\ d & e & f \end{smallmatrix} \right) \geq \frac{1}{(k+1)^{3(n-1)}}, \quad (144)$$

where $(k+1)^{3(n-1)}$ is an upper bound on the number of triples (α, β, δ) such that $\varphi_\alpha(a) \varphi_\beta(b) \varphi_\delta(d) \neq 0$ as there are at most $(k+1)^{\ell-1}$ Young diagrams with at most k boxes and at most ℓ rows.

Our objective now is to derive a complementary upper bound on $W_k \left(\begin{smallmatrix} a & b & c \\ d & e & f \end{smallmatrix} \right)$. To accomplish this, we note that an immediate corollary of Corollary 51 is the following bound which holds for any triple (x, y, z) of positive semidefinite eigenvalues:

$$\left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_2^2 \varphi_\alpha(x) \varphi_\beta(y) \varphi_\delta(z) \leq R_\epsilon^{\alpha\beta\delta} \max_{\substack{(X,Y,Z) \\ \in \mathcal{O}_x \times \mathcal{O}_y \times \mathcal{O}_z}} \left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \right|, \quad (145)$$

where $R_\epsilon^{\alpha\beta\delta} = \frac{\dim(V_\alpha^n \otimes V_\beta^n \otimes V_\delta^n)}{\dim(V_\epsilon^n)}$ and the where the maximum is taken over all triples (X, Y, Z) of positive semidefinite matrices with eigenvalues (x, y, z) . Applying Theorem 45 (and its related corollaries) to $\left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \right|$ on the right-hand side above produces

$$\left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \right| \leq \left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_\infty \varphi_\phi(Y+Z)^{\frac{1}{3}} \varphi_\gamma(X+Y)^{\frac{1}{3}} \varphi_\epsilon(X+Y+Z)^{\frac{1}{3}}. \quad (146)$$

Then Eq. (146) can be applied to each summand in $W_k \left(\begin{smallmatrix} a & b & c \\ d & e & f \end{smallmatrix} \right)$ (replacing (x, y, z) with (a, b, d)) to obtain

$$W_k \left(\begin{smallmatrix} a & b & c \\ d & e & f \end{smallmatrix} \right) \leq \left(\frac{k!}{\text{Tr}[e]^k} \right)^2 \sum_{\substack{\alpha, \beta, \delta \\ \gamma, \epsilon, \phi \\ |\epsilon| = k}} R_\epsilon^{\alpha\beta\delta} \varphi_\alpha(a)^{\frac{2}{3}} \varphi_\delta(d)^{\frac{2}{3}} \max_{\substack{(X,Y,Z) \\ \in \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_d}} [\varphi_\phi(f) \varphi_\phi(Y+Z) \varphi_\gamma(c) \varphi_\gamma(X+Y) \varphi_\epsilon(e) \varphi_\epsilon(X+Y+Z)]^{\frac{1}{3}}. \quad (147)$$

Note that here we have made use of the inequality $\left\| \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\|_2^2 \geq \left\| \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\|_\infty^2$ to simplify the presentation slightly. Now the eigenvalue separation lemma implies a triple of bounds (because $|\alpha|+|\phi| = |\gamma|+|\delta| = |\epsilon| = k$)

$$\varphi_\alpha(a) \sqrt{\varphi_\phi(f) \varphi_\phi(Y+Z)} \leq \dim(V_\alpha^n \otimes V_\phi^n) \frac{\text{Tr}[e]^k}{k!} \exp\left(-\frac{k}{4 \text{Tr}[e]^2} \|f - \text{eig}(Y+Z)\|_1^2\right), \quad (148)$$

$$\varphi_\delta(d) \sqrt{\varphi_\gamma(c) \varphi_\gamma(X+Y)} \leq \dim(V_\gamma^n \otimes V_\delta^n) \frac{\text{Tr}[e]^k}{k!} \exp\left(-\frac{k}{4 \text{Tr}[e]^2} \|c - \text{eig}(X+Y)\|_1^2\right), \quad (149)$$

$$\sqrt{\varphi_\epsilon(e) \varphi_\epsilon(X+Y+Z)} \leq \dim(V_\epsilon^n) \frac{\text{Tr}[e]^k}{k!} \exp\left(-\frac{k}{4 \text{Tr}[e]^2} \|e - \text{eig}(X+Y+Z)\|_1^2\right). \quad (150)$$

Taking the product of these three inequalities to the power of $2/3$, applying it to the upper bound on W_k and combining it with the lower bound on W_k then implies

$$\min_{\substack{(X,Y,Z) \\ \in \mathcal{O}_a \times \mathcal{O}_b \times \mathcal{O}_d}} \{\|f - \text{eig}(Y+Z)\|_1^2 + \|c - \text{eig}(X+Y)\|_1^2 + \|e - \text{eig}(X+Y+Z)\|_1^2\} \leq \frac{6 \text{Tr}[e]^2}{k} (3n(n-1) + 9(n-1)) \log(k+1). \quad (151)$$

where the $6(n-1)$ factor is an upper bound on the number of non-zero summands for the right-hand side, and additional factor of $3(n-1)$ comes from the lower bound on W_k and where the $3n(n-1)$ factor comes from an upper bound to the dimensions:

$$\frac{\dim(V_\alpha^n \otimes V_\beta^n \otimes V_\delta^n)}{\dim(V_\epsilon^n)} \dim(V_\alpha^n \otimes V_\phi^n \otimes V_\gamma^n \otimes V_\delta^n \otimes V_\epsilon^n)^{\frac{2}{3}} \leq \left[(k+1)^{\frac{n(n-1)}{2}}\right]^{3+4\frac{2}{3}} \leq (k+1)^{3n(n-1)}. \quad (152)$$

Therefore, the deviation measure is bounded by

$$D\left(\begin{smallmatrix} a & b & c \\ d & e & f \end{smallmatrix}\right) \leq \text{Tr}[e] \sqrt{(18n^2 + 36n - 54) \frac{\log(k+1)}{k}} \quad (153)$$

which proves the claim. $\square$

3.10 Proof of Theorem 5

In order to prove Theorem 5 it will be useful to have a bound on the quantity $\left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \right|$ which directly relates the sextuple of Young diagrams $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ to the eigenvalues of $(X, Y, X+Y, Z, X+Y+Z, Y+Z)$. lem:asym_size_helper

Lemma 53. *To begin let $X, Y, Z \geq 0$ be positive semidefinite $n \times n$ matrices and let (a, b, c, d, e, f) be the eigenvalues of the matrices $(X, Y, X+Y, Z, X+Y+Z, Y+Z)$. Then for all sextuples of Young diagrams, $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$, with sizes satisfying*

$$k = |\epsilon| = |\alpha| + |\beta| + |\delta| = |\alpha| + |\phi| = |\gamma| + |\delta|, \quad (154) \quad \text{\small \{eq:yd_k_sizes\}}$$

we have (assuming $\text{Tr}[e] \neq 0$)

$$\frac{k!}{\text{Tr}[e]^k} \left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \right| \leq \min\left\{ \left\| \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\|_\infty, (k+1)^{n(n-1)} \exp\left(-\frac{k}{4} R_{abcdef}^{\alpha\beta\gamma\delta\epsilon\phi}\right) \right\}, \quad (155)$$

where

$$R_{abcdef}^{\alpha\beta\gamma\delta\epsilon\phi} = D\left(\frac{\alpha \oplus \beta \oplus \delta}{k} \left\| \frac{a \oplus b \oplus d}{\text{Tr}[e]} \right.\right) + D\left(\frac{\gamma \oplus \delta}{k} \left\| \frac{c \oplus d}{\text{Tr}[e]} \right.\right) + D\left(\frac{\alpha \oplus \phi}{k} \left\| \frac{a \oplus f}{\text{Tr}[e]} \right.\right) + D\left(\frac{\epsilon}{k} \left\| \frac{e}{\text{Tr}[e]} \right.\right). \quad (156)$$

Proof. A combination of Theorem 45 and Lemma 46 implies

$$\left| \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(X, Y, Z) \right| \leq \left\| \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\|_\infty (\varphi_\alpha(a) \varphi_\beta(b) \varphi_\delta(d) \times \varphi_\gamma(c) \varphi_\delta(d) \times \varphi_\epsilon(e) \times \varphi_\alpha(a) \varphi_\phi(f))^{\frac{1}{4}}, \quad (157)$$

Additionally, the relative entropy bounds arising from the multinomial distribution places the following bounds on four probability distributions:

$$\frac{k!}{\text{Tr}[e]^k} \varphi_\alpha(a) \varphi_\beta(b) \varphi_\delta(d) \leq \dim(V_\alpha^n \otimes V_\beta^n \otimes V_\delta^n) \exp \left[-kD \left(\frac{\alpha \oplus \beta \oplus \delta}{k} \left\| \frac{a \oplus b \oplus d}{\text{Tr}[e]} \right\| \right) \right], \quad (158)$$

$$\frac{k!}{\text{Tr}[e]^k} \varphi_\gamma(c) \varphi_\delta(d) \leq \dim(V_\gamma^n \otimes V_\delta^n) \exp \left[-kD \left(\frac{\gamma \oplus \delta}{k} \left\| \frac{c \oplus d}{\text{Tr}[e]} \right\| \right) \right], \quad (159)$$

$$\frac{k!}{\text{Tr}[e]^k} \varphi_\alpha(a) \varphi_\phi(f) \leq \dim(V_\alpha^n \otimes V_\phi^n) \exp \left[-kD \left(\frac{\alpha \oplus \phi}{k} \left\| \frac{a \oplus f}{\text{Tr}[e]} \right\| \right) \right], \quad (160)$$

$$\frac{k!}{\text{Tr}[e]^k} \varphi_\epsilon(e) \leq \dim V_\epsilon^n \exp \left[-kD \left(\frac{\epsilon}{k} \left\| \frac{e}{\text{Tr}[e]} \right\| \right) \right]. \quad (161)$$

Since the product of the four distributions on the left-hand side above is at most the minimum of one or the product of relative entropy bounds on the right-hand side above (and the product of the dimensions to the fourth root is at most $[(k+1)^{\frac{n(n-1)}{2}}]^{\frac{8}{4}}$), obtain the claim. $\square$

Proof of Theorem 5. To begin we use Lemma 53 to prove the claimed inverse polynomial lower bound. Assume (a, b, c, d, e, f) are the eigenvalues of positive semidefinite matrices (A, B, C, D, E, F) satisfying the tetrahedral constraints. Summing $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(A, B, D)$ over all sextuples of Young diagrams satisfying Eq. (154) produces

$$\sum_{\substack{\alpha, \beta, \delta \\ \gamma, \epsilon, \phi \\ |\epsilon|=k}} \varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(A, B, D) = \frac{\text{Tr}[e]^k}{k!}. \quad (162)$$

Therefore, the sextuple which maximizes the absolute value of $\varphi_{\gamma\epsilon\phi}^{\alpha\beta\delta}(A, B, D)$ over this k -dependent domain of summation, which we denote by $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$, must satisfy

$$\frac{k!}{\text{Tr}[e]^k} \left| \varphi_{\gamma_k \epsilon_k \phi_k}^{\alpha_k \beta_k \delta_k}(A, B, D) \right| \geq \frac{1}{(k+1)^{6(\text{rank}[e]-1)}} \quad (163)$$

where $(k+1)^{6(\text{rank}[e]-1)}$ is a crude upper bound on the number of terms in the above summation which are non-zero as there are at most $(k+1)^{r-1}$ Young diagrams of size at most k and depth at most r (note that each of $\varphi_\alpha(a), \varphi_\beta(b), \varphi_\delta(d), \varphi_\phi(f), \varphi_\gamma(c)$ and $\varphi_\epsilon(e)$ are respectively only non-zero if the depth of the corresponding Young diagram is at most the rank of the corresponding set of eigenvalues and $\text{rank}[e]$ is always the maximum possible rank). Therefore, Lemma 53 immediately implies both

$$\left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|_\infty \geq \frac{1}{(k+1)^{6(\text{rank}[e]-1)}}, \quad \text{\texttt{eq:n_independent_inverse_poly}} \quad (164)$$

as well as

$$\exp\left(-\frac{k}{4} R_{abcdef}^{\alpha_k \beta_k \gamma_k \delta_k \epsilon_k \phi_k}\right) \geq \frac{1}{(k+1)^{n(n-1)}} \frac{1}{(k+1)^{6(\text{rank}[e]-1)}}. \quad (165)$$

Thus we have demonstrated the existence of a sequence of sextuples of Young diagrams for which (i) $\left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|_\infty$ satisfies the claimed inverse polynomial lower bound (which is actually independent of n), and (ii) $R_{abcdef}^{\alpha_k \beta_k \gamma_k \delta_k \epsilon_k \phi_k} \rightarrow 0$ which implies the convergence claim:

$$\frac{\alpha_k}{k} \rightarrow \frac{a}{\text{Tr}[e]}, \quad \frac{\beta_k}{k} \rightarrow \frac{b}{\text{Tr}[e]}, \quad \frac{\gamma_k}{k} \rightarrow \frac{c}{\text{Tr}[e]}, \quad \frac{\delta_k}{k} \rightarrow \frac{d}{\text{Tr}[e]}, \quad \frac{\epsilon_k}{k} \rightarrow \frac{e}{\text{Tr}[e]}, \quad \frac{\phi_k}{k} \rightarrow \frac{f}{\text{Tr}[e]}. \quad (166)$$

To prove the exponential decay on $\left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_\infty$ the case where (a, b, c, d, e, f) are not tetrahedral, it will be slightly easier to prove the exponential decay of $\left\| \begin{Bmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{Bmatrix} \right\|_2$ which is actually a bit stronger than necessary. Let (a, b, c, d, e, f) be the set of non-tetrahedral eigenvalues to which the sequence of Young diagrams

$(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$ is converging in proportion. We can henceforth assume the Young diagrams satisfy the size constraints Eq. (154) as before. Then from Corollary 51, we conclude

$$\left\| \left\{ \begin{smallmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{smallmatrix} \right\} \right\|_2^2 \leq \max_{\substack{X,Y,Z \\ \text{Tr}(X)=\text{Tr}[a] \\ \text{Tr}(Y)=\text{Tr}[b] \\ \text{Tr}(Z)=\text{Tr}[d]}} (k+1)^{\frac{3n^2(n^2-1)}{2}} \frac{|\alpha_k|! |\beta_k|! |\delta_k|!}{\text{Tr}[a]^{|\alpha_k|} \text{Tr}[b]^{|\beta_k|} \text{Tr}[d]^{|\delta_k|}} \left| \varphi_{\gamma_k \epsilon_k \phi_k}^{\alpha_k \beta_k \delta_k}(X, Y, Z) \right|. \quad (167)$$

Now because $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$ are converging in proportion to the eigenvalues (a, b, c, d, e, f) we can assume that the sizes are such that the following multinomial distribution bound holds:

$$\frac{|\epsilon_k|!}{\text{Tr}[e]^{|\epsilon_k|}} \frac{\text{Tr}[a]^{|\alpha_k|} \text{Tr}[b]^{|\beta_k|} \text{Tr}[d]^{|\delta_k|}}{|\alpha_k|! |\beta_k|! |\delta_k|!} \geq \frac{1}{\text{poly}(k)}. \quad (168)$$

Using Lemma 53 then gives

$$\left\| \left\{ \begin{smallmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{smallmatrix} \right\} \right\|_2^2 \leq \text{poly}(k) \exp\left(-\frac{k}{4} \min R_{a'b'c'd'e'f'}^{\alpha_k \beta_k \gamma_k \delta_k \epsilon_k \phi_k}\right). \quad (169)$$

where the minimum is taken over all tetrahedral eigenvalues (a', b', c', d', e', f') with matching traces. However, since $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$ are converging to non-tetrahedral eigenvalues (a, b, c, d, e, f) , there must exist a sufficiently large K such that for all $k \geq K$, $\min R_{a'b'c'd'e'f'}^{\alpha_k \beta_k \gamma_k \delta_k \epsilon_k \phi_k} \geq \delta$ for some positive constant $\delta > 0$ which proves the claim. $\square$

3.11 The large n limit & an entropic inequality

sec:large_n

The purpose of this section is to show how to leverage the inverse polynomial bound on $\left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|$ for tetrahedral (a, b, c, d, e, f) provided by Theorem 5 to derive inequalities on the entropies of the eigenvalues. Our strategy will be to prove an upper bound on the $6j$ symbol which depends only on the dimensions of the underlying irreducible representations and the multiplicity spaces between them. By combining this upper bound with the lower bound provided by Theorem 5 (specifically the n -independent bound in Eq. (164)), we will prove the following.

thm:entropic_ineq

Theorem 54. *Let (a, b, c, d, e, f) be tetrahedral eigenvalues. Then*

$$s(c) + s(f) - s(b) - s(e) \geq 0, \quad (170)$$

eq:entropic_ineq

where

$$s(x) := - \sum_{i=1}^n x_i \log x_i. \quad (171)$$

eq:little_s_defn

We begin with our upper bound on the $6j$ symbol.

lem:sixj_ssa_bound

Lemma 55. *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be a sextuple of Young diagrams. Then*

$$\left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_2^2 \leq \left(c_{\alpha, \phi}^{\epsilon} c_{\beta, \delta}^{\phi} c_{\alpha, \beta}^{\gamma} c_{\gamma, \delta}^{\epsilon} \right)^{\frac{1}{2}} \frac{\dim(V_{\gamma}^n) \dim(V_{\phi}^n)}{\dim(V_{\beta}^n) \dim(V_{\epsilon}^n)}. \quad (172)$$

Proof. Here we provide a diagrammatic proof of the claimed inequality. For the sake of clarity, we abbreviate the irreducible representation labels, V_{λ}^n , and omit the multiplicity space labels on the intermediate steps.

$$\left(\begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right)^{\dagger} \left(\begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right) = \left[\begin{smallmatrix} \gamma & \delta & \epsilon \\ \alpha & \beta & \phi \end{smallmatrix} \right] \left[\begin{smallmatrix} \gamma & \delta & \epsilon \\ \alpha & \beta & \phi \end{smallmatrix} \right] \leq \left[\begin{smallmatrix} \gamma & \delta & \epsilon \\ \alpha & \beta & \phi \end{smallmatrix} \right] \left[\begin{smallmatrix} \gamma & \delta & \epsilon \\ \alpha & \beta & \phi \end{smallmatrix} \right]^{\frac{1}{2}} = \frac{\epsilon^2 \gamma \phi}{\beta \epsilon} \left[\begin{smallmatrix} \epsilon & \gamma & \phi \\ \alpha & \beta & \delta \end{smallmatrix} \right]^{\frac{1}{2}} \quad (173)$$

The first equality follows from Definition 41. The inequality follows from an application of the Cauchy-Schwarz inequality with respect to the annotated contraction. The final equality follows from applying Schur's lemma to components of the diagram mapping V_{β}^n to itself on the left and V_{ϵ}^n to itself on the right. Note that the factors of $\dim(V_{\epsilon}^n)^2$ appearing on left-most and right-most sides of the above inequality cancel producing the claimed inequality. $\square$

An immediate application of Lemma 55 is to combine it with Eq. (164). If (a, b, c, d, e, f) are tetrahedral eigenvalues, then by Theorem 5 there exists a converging sequence of Young diagrams $(\alpha_k, \beta_k, \gamma_k, \delta_k, \epsilon_k, \phi_k)$ such that

$$\frac{1}{q(k)} \leq \left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|_2^2 \leq \left(c_{\alpha_k, \phi_k}^{\epsilon_k} c_{\beta_k, \delta_k}^{\phi_k} c_{\alpha_k, \beta_k}^{\gamma_k} c_{\gamma_k, \delta_k}^{\epsilon_k} \right)^{\frac{1}{2}} \frac{\dim(V_{\gamma_k}^n) \dim(V_{\phi_k}^n)}{\dim(V_{\beta_k}^n) \dim(V_{\epsilon_k}^n)}. \quad \text{\texttt{eq:combined_bound}} \quad (174)$$

where $q(k)$ is some polynomial in k independent of the dimension n (see Eq. (164)). Remarkably, the Littlewood-Richardson coefficients $c_{\alpha_k, \phi_k}^{\epsilon_k}$, $c_{\beta_k, \delta_k}^{\phi_k}$, $c_{\alpha_k, \beta_k}^{\gamma_k}$, and $c_{\gamma_k, \delta_k}^{\epsilon_k}$ are also *independent* of the dimension parameter n defining the group $U(n)$ (assuming n is at least as large as the maximum depth of any of the involved Young diagrams. This observation provides us with an opportunity to optimize the above bound by varying the value of n .

In particular, it will be useful to better understand how the dimension, $\dim(V_{\lambda}^n)$, of the irreducible representation V_{λ}^n of $U(n)$ grows as n becomes much larger than the number of rows of λ . In general, for fixed λ , the function $n \mapsto V_{\lambda}^n$ can grow at an exponential rate with increasing n . One can find estimates for $\dim(V_{\lambda}^n)$ in the regime where n is large compared to $|\lambda|$ and also the last row $\lambda_{\min}$ is large compared to the depth of λ in paper by Itzykson and Nauenberg [IN66, Section III]. Here we use an elegant presentation of this scaling, which relates the asymptotics of $\dim(V_{\lambda}^n)$ to the value of $\dim(W_{\lambda})$ (see [Har23, Lemma 2] or [CHW07, Equations (25, 26)]).

prop:harrow_formula

Proposition 56. *Let λ be a Young diagram with depth at most n . Then*

$$\frac{\dim(V_{\lambda}^n)}{n^{|\lambda|}} = \frac{\dim(W_{\lambda})}{|\lambda|!} \prod_{(i,j) \in \lambda} \left(1 + \frac{j-i}{n} \right). \quad (175)$$

Proof. The statement follows from the dimension formulas for $\dim(V_{\lambda}^n)$ and $\dim(W_{\lambda})$ in Eq. (50) and Eq. (53). See also [Har23, Appendix A]. $\square$

From Proposition 56 we obtain the following corollary.

cor:harrow_ssa_lim

Corollary 57. *Let $\gamma, \phi, \beta, \epsilon$ be a quadruple of Young diagrams such that*

$$|\gamma| + |\phi| = |\beta| + |\epsilon|. \quad \text{\texttt{eq:balanced_sizes}} \quad (176)$$

Then

$$\lim_{n \rightarrow \infty} \frac{\dim(V_{\gamma}^n) \dim(V_{\phi}^n)}{\dim(V_{\beta}^n) \dim(V_{\epsilon}^n)} = \frac{|\beta|! |\epsilon|! \dim(W_{\gamma}) \dim(W_{\phi})}{|\gamma|! |\phi|! \dim(W_{\beta}) \dim(W_{\epsilon})} = \frac{H_{\gamma} H_{\phi}}{H_{\beta} H_{\epsilon}} \quad (177)$$

Proof. From Proposition 56 one evidently has for any Young diagram λ the limit

$$\lim_{n \rightarrow \infty} \frac{\dim(V_{\lambda}^n)}{n^{|\lambda|}} = \frac{\dim(W_{\lambda})}{|\lambda|!} = \frac{1}{H_{\lambda}}. \quad (178)$$

The claim then follows because the missing prefactor, $\frac{n^{|\beta|} n^{|\epsilon|}}{n^{|\gamma|} n^{|\phi|}}$, is equal to one by Eq. (176). $\square$

We are now in a position to prove the main result of this section.

Proof of Theorem 54. As (a, b, c, d, e, f) are tetrahedral, the inverse polynomial bound $\left\| \begin{Bmatrix} \alpha_k & \beta_k & \gamma_k \\ \delta_k & \epsilon_k & \phi_k \end{Bmatrix} \right\|$ implies it is non-zero and thus Eq. (176) holds. Therefore, by Corollary 57 and the discussion surrounding the bound in Eq. (174) we conclude that along the convergent sequence of Young diagrams,

$$\lim_{k \rightarrow \infty} \frac{1}{k} \log \left[\frac{H_{\beta_k} H_{\epsilon_k}}{H_{\gamma_k} H_{\phi_k}} \right] \geq 0 \quad \text{\texttt{eq:hook_ratio_limit}} \quad (179)$$

provided the limit. To prove the limit exists, an application of Stirling's approximation, $\log n! \approx n \log n - n$ [Rob55], to the approximation formula $H_\lambda \approx \prod_{i=1}^n \lambda_i!$ produces for any constant η the formula

$$\frac{1}{k} \log H_\lambda \approx (\log \eta - 1)|\lambda| + \sum_{i=1}^n \frac{\lambda_i}{\eta} \log \frac{\lambda_i}{\eta} \quad (180)$$

Because $|\gamma| + |\phi| = |\beta| + |\epsilon|$ (Eq. (176)), the $(\log \eta - 1)$ terms will cancel in Eq. (179). Then using the fact that that $(\beta_k, \gamma_k, \epsilon_k, \phi_k)$ converge to proportion to the eigenvalues (b, c, e, f) (with proportionality constant $\eta = \frac{k}{\text{Tr}[e]}$), we conclude that Eq. (179) has a limit can indeed equals the left-hand side of Eq. (170). $\square$

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